

Georeferencing Overview

University of Florida

Sarah Ellen Strickland

UF | UNIVERSITY of FLORIDA

FLORIDA MUSEUM™

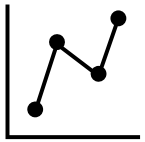
iDigBio
Integrated Digitized Biocollections

NSF

Agenda



What is a georeference?



Why georeference?



Key Concepts



How do you georeference?

What is georeferencing?



What is georeferencing?



FLORA OF IDAHO

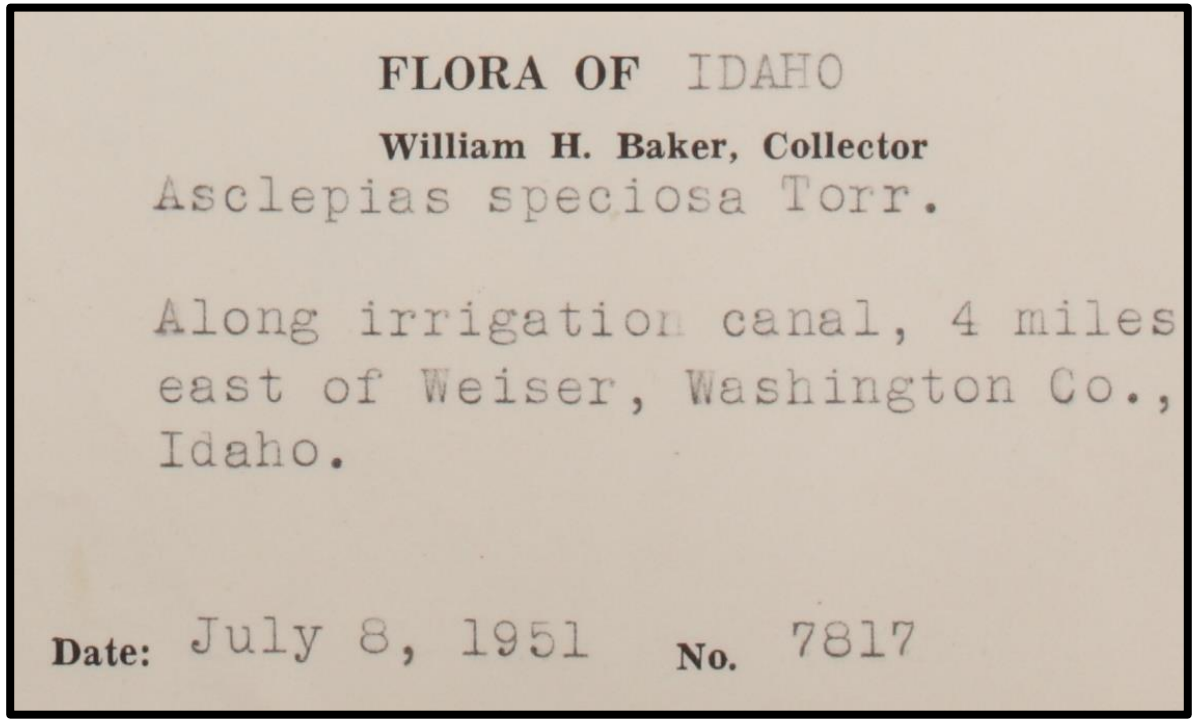
William H. Baker, Collector

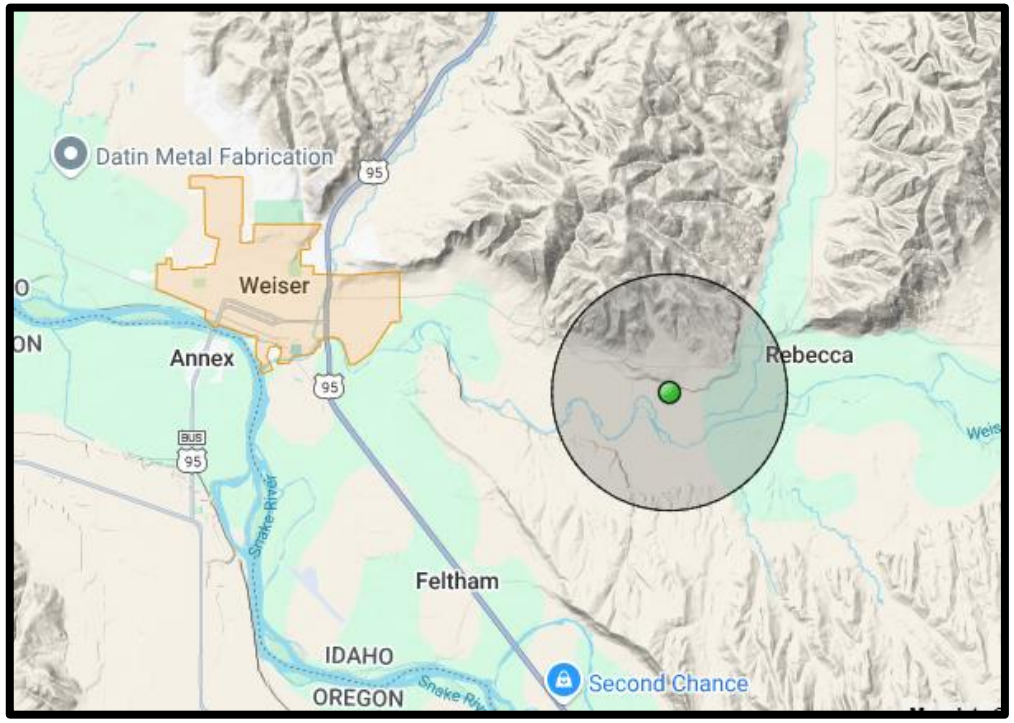
Asclepias speciosa Torr.

Along irrigation canal, 4 miles east of Weiser, Washington Co., Idaho.

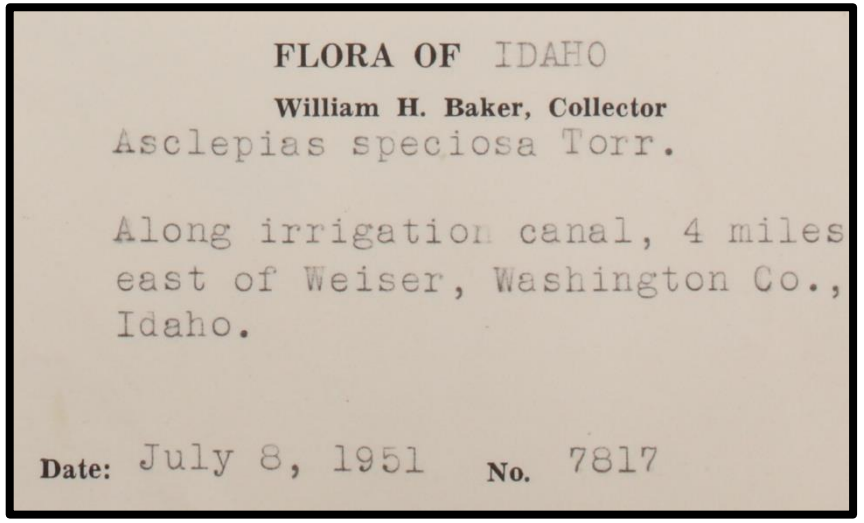
Date: July 8, 1951 **No.** 7817

What is georeferencing?





What is georeferencing?



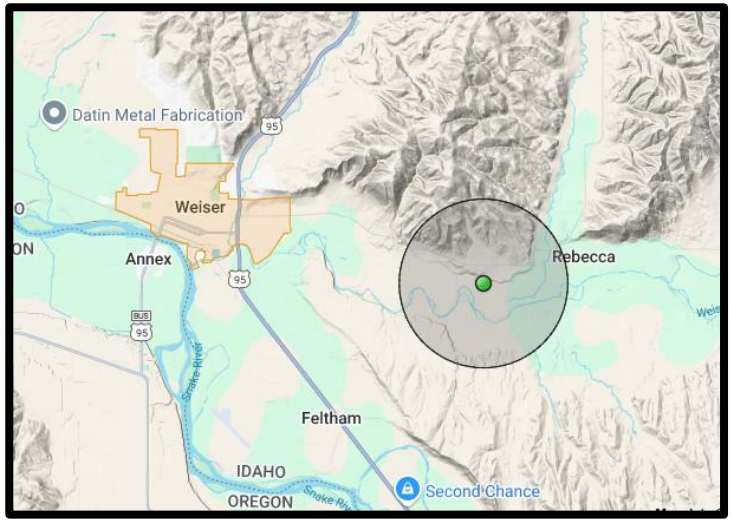
FLORA OF IDAHO

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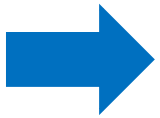
44.23, -116.88

Uncertainty: 1956 m

What is a georeference?

***Asclepias speciosa* Torr.**

Along irrigation canal, 4 miles east of Weiser, Washington Co., Idaho.



44.23, -116.88

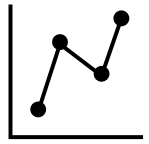
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A numerical description of a place that can be mapped

Agenda



What is a georeference?



Why georeference?



Key Concepts



How do you georeference?

More occurrence data







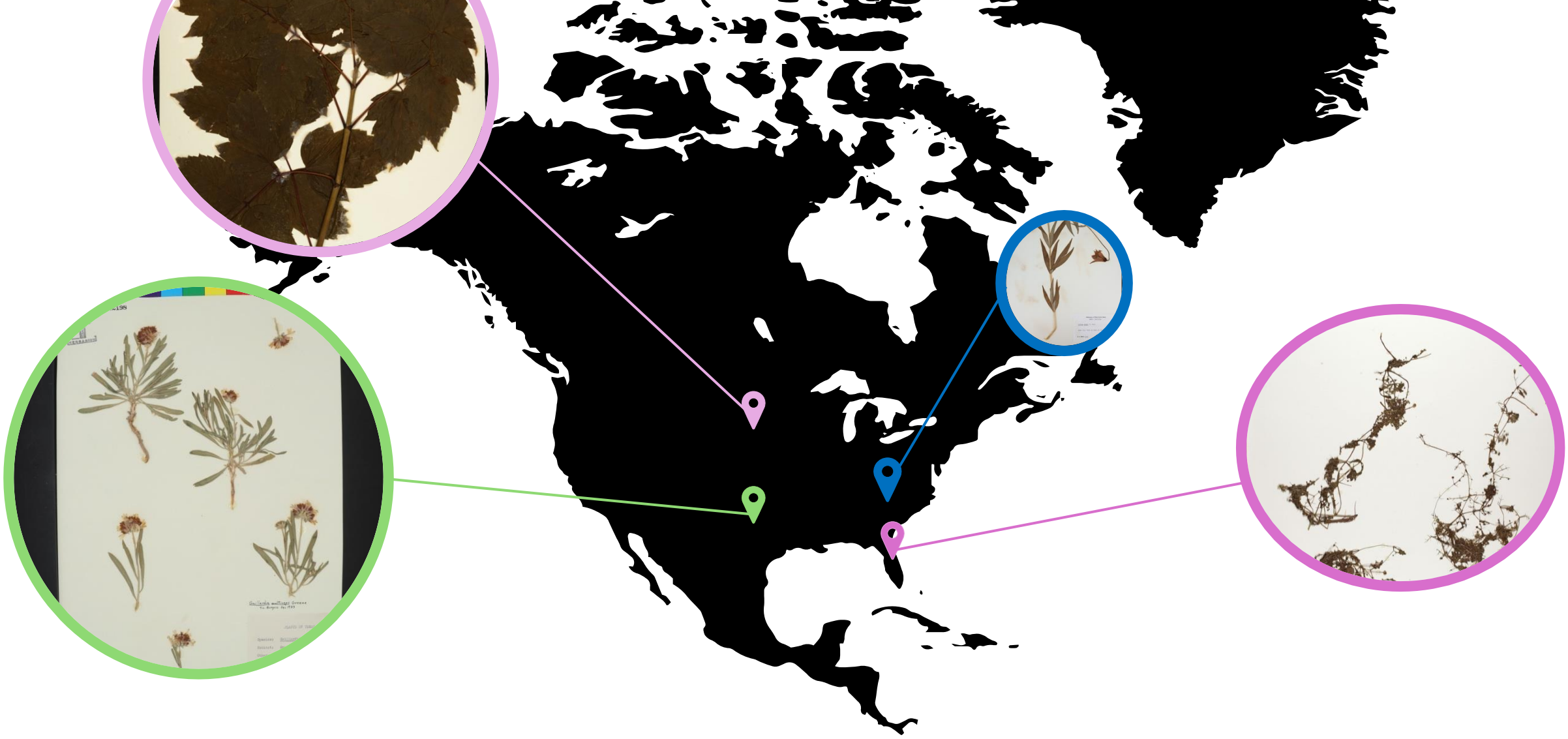




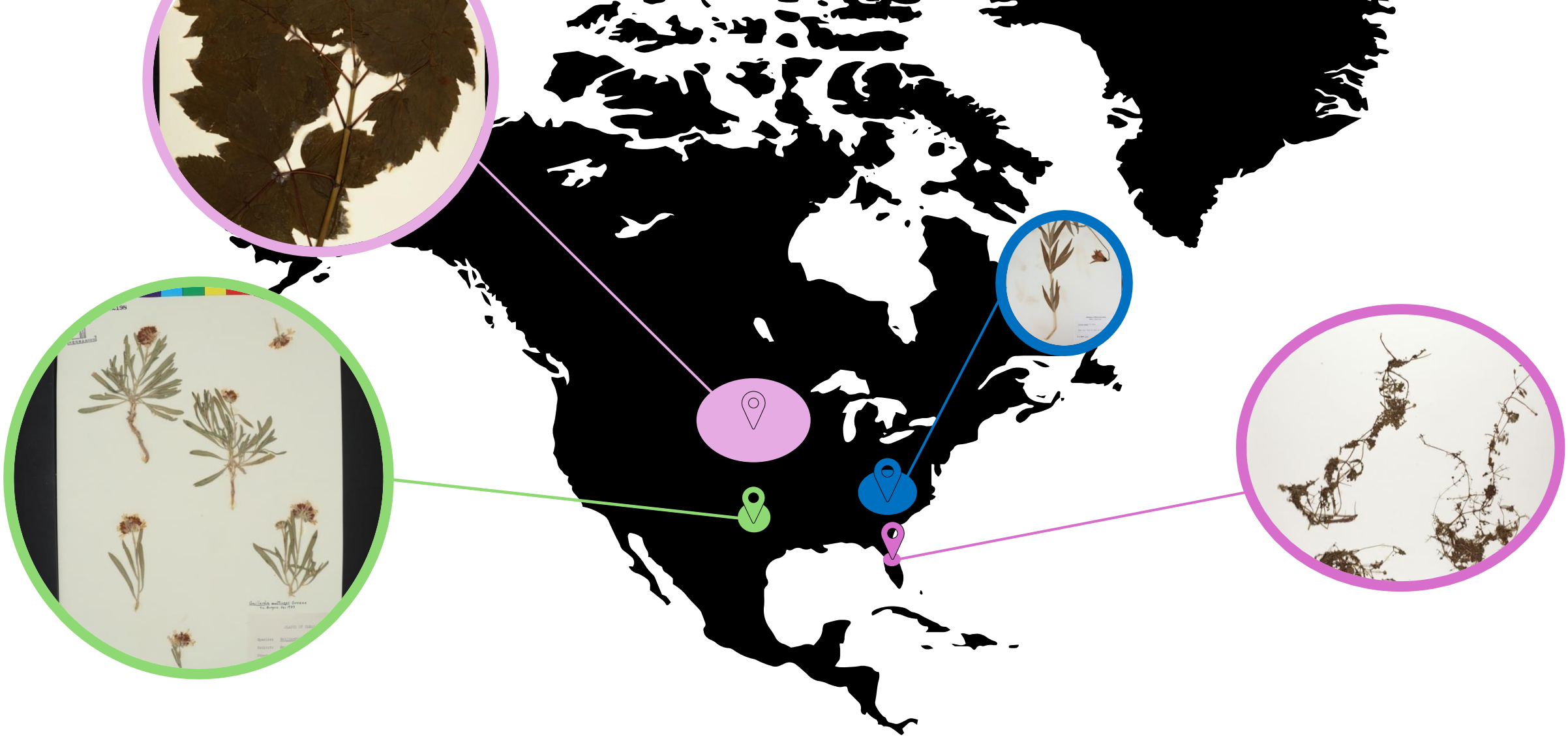








More occurrence data



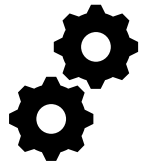
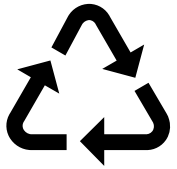
Filter by uncertainty

Additional reliable data

Why georeference?

Findable **Accessible** **Interoperable** **Reusable**



***Asclepias speciosa* Torr.**

Along irrigation canal, 4 miles east of Weiser, Washington Co., Idaho.



Latitude:	Longitude:	Uncertainty (m):
44.23	-116.88	1956

Agenda



What is a georeference?



Why georeference?

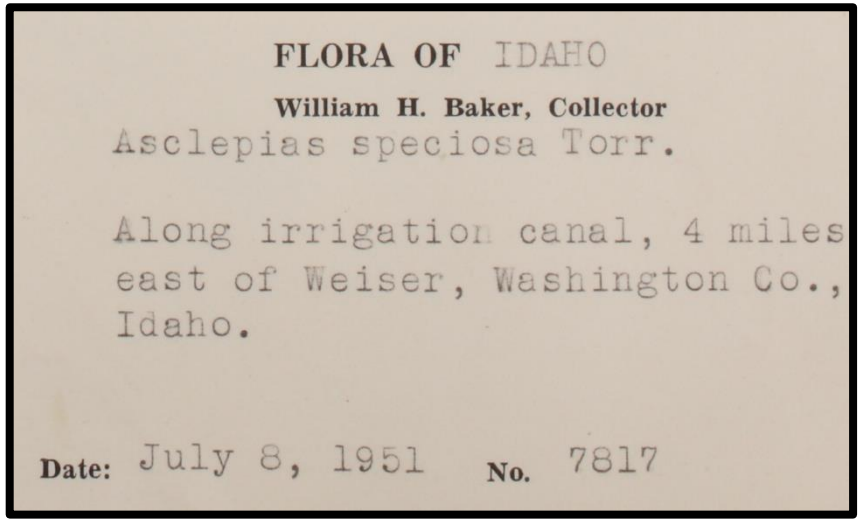


Key Concepts



How do you georeference?

What is georeferencing?



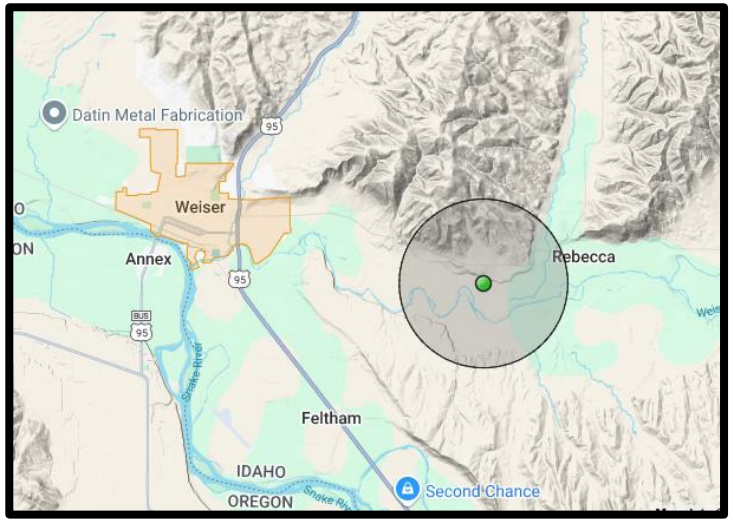
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Uncertainty: 1956 m

A numerical description of a place that can be mapped

Coordinates



44.23, -116.88

Uncertainty



1956 m

Coordinate Formats

Degrees minutes and seconds (DMS)	Decimal degrees (DD)	Degrees and decimal minutes (DDM)
36º 31' 21.4" N 114º 09' 50.6" W	−36.524276, −114.164055	36º 31.4566’N 114º 09.8433’W

Coordinate Formats

RESET VALUES

Decimal Degrees (DD)

Latitude:

-36.524276 °

Longitude:

-114.164055 °

Latitude (-90 to 90) and longitude (-180 to 180). Include up to 6 decimal places.

Degrees Decimal Minutes (DDM)

Latitude:

36 ° 31.4566 ' S

Longitude:

114 ° 9.8433 ' W

Degrees (0 to 89, 0 to 179) as integers and minutes (0 to 59.9999)
Include up to 4 decimal places.

Degrees Minutes Seconds (DMS)

Latitude:

36 ° 31 ' 27.3936 " S

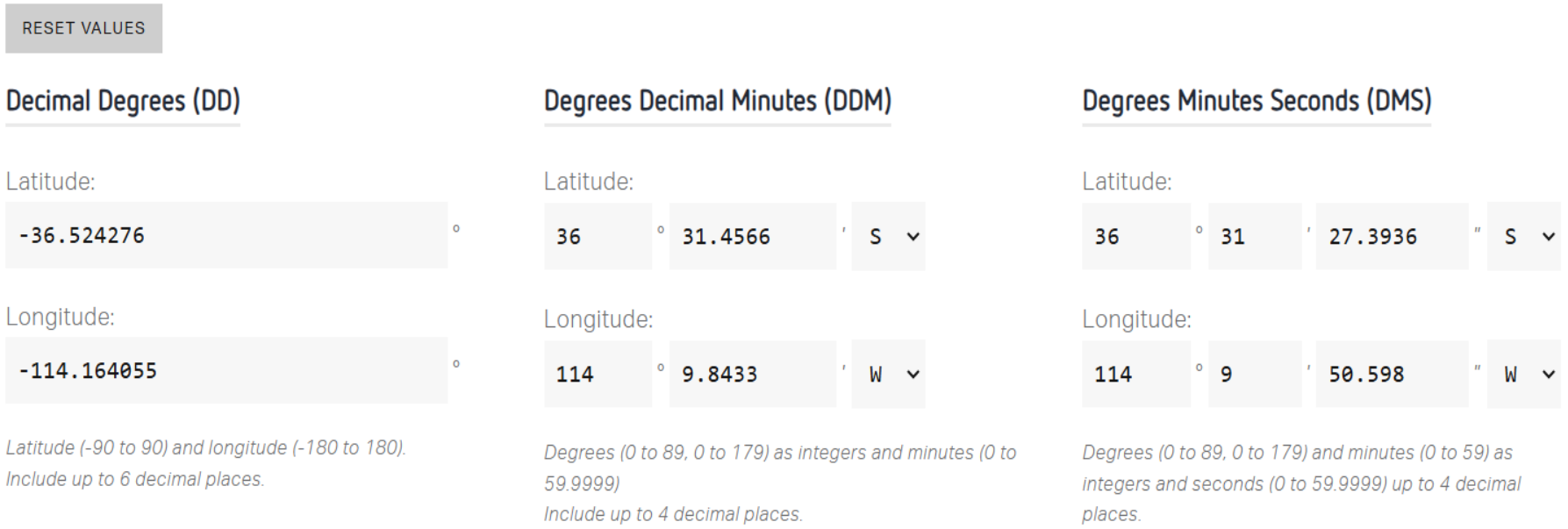
Longitude:

114 ° 9 ' 50.598 " W

Degrees (0 to 89, 0 to 179) and minutes (0 to 59) as integers and seconds (0 to 59.9999) up to 4 decimal places.

https://applications.pgc.umn.edu/convert/

Coordinate Formats



https://applications.pgc.umn.edu/convert/

Batch Degree Conversion: https://blog.batchgeo.com/convert-latitude-longitude-to-decimals-excel/

A numerical description of a place that can be mapped

Coordinates



44.23, -116.88

Uncertainty



1956 m

Uncertainty

Every location has associated uncertainty

coordinateUncertaintyInMeters

There are MANY different sources

Locality Description

Distance measurement precision

Direction measurement precision

Point Location

Locality extent

Coordinate Precision and GPS accuracy

Mapping

Map scale and Coordinate Reference Systems

Uncertainty in Locality Descriptions

“3 miles north of Tucson”

Uncertainty in Locality Descriptions

"**3 miles** north of Tucson"

3.0 miles?

2.75 miles?

3.03 miles?

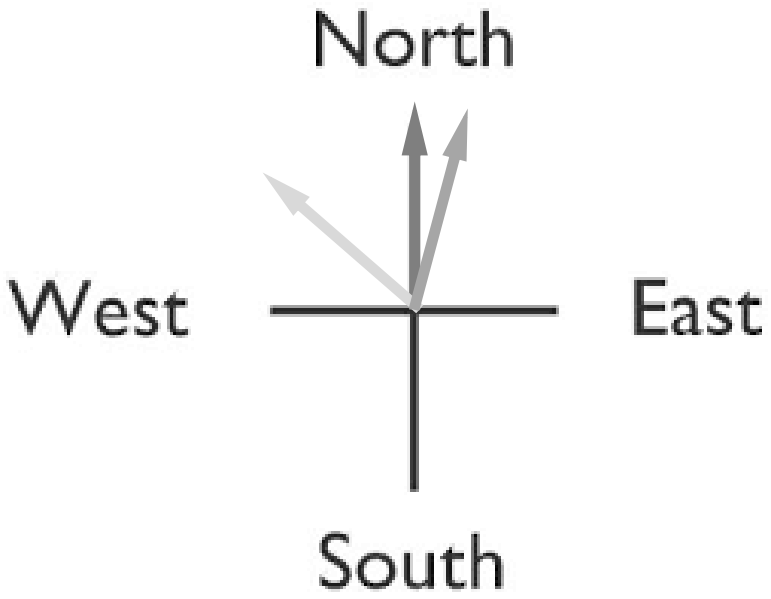
Uncertainty in Locality Descriptions

"3 miles **north** of Tucson"

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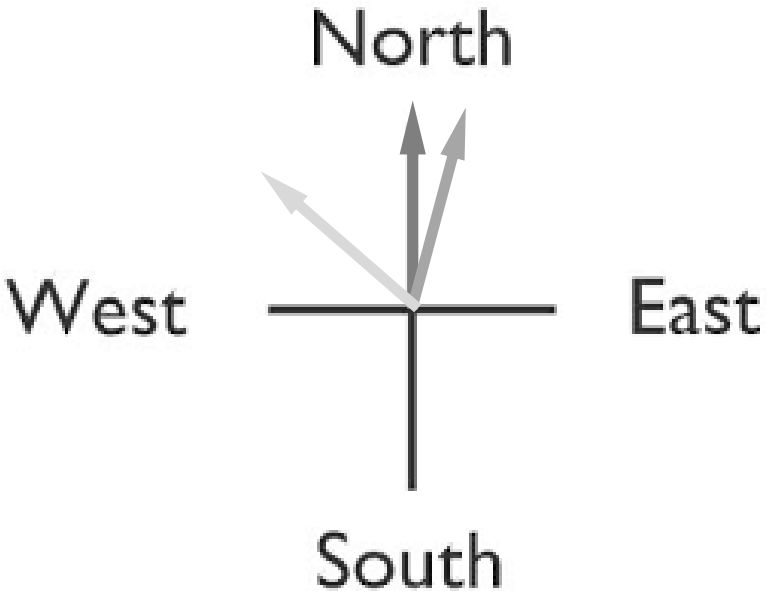
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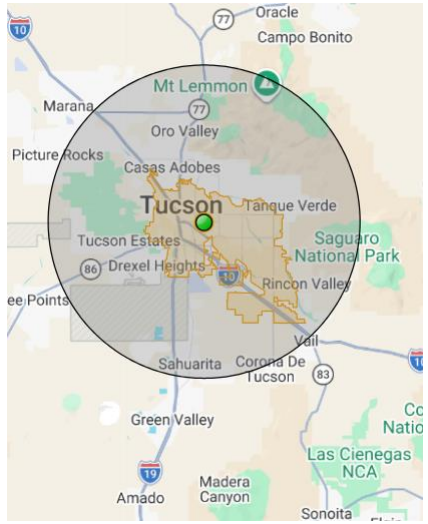
"3 miles north of **Tucson**"

3.0 miles?

2.75 miles?

3.03 miles?





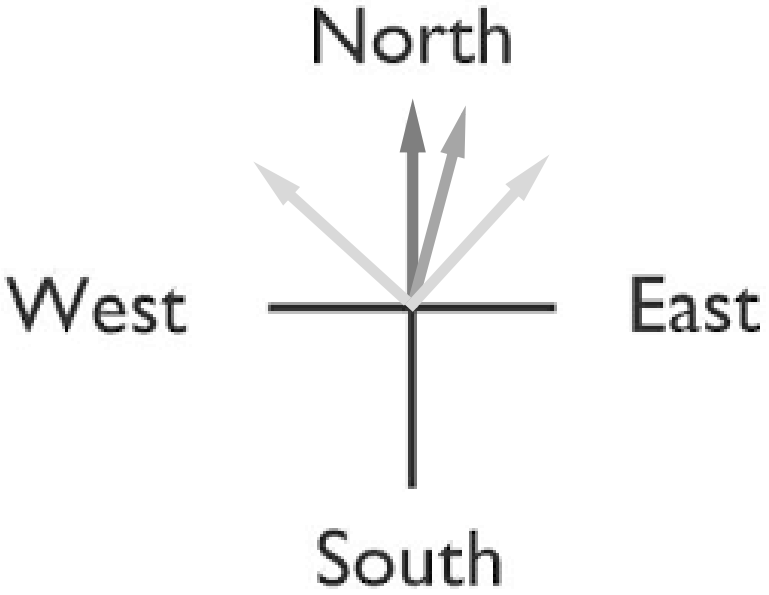
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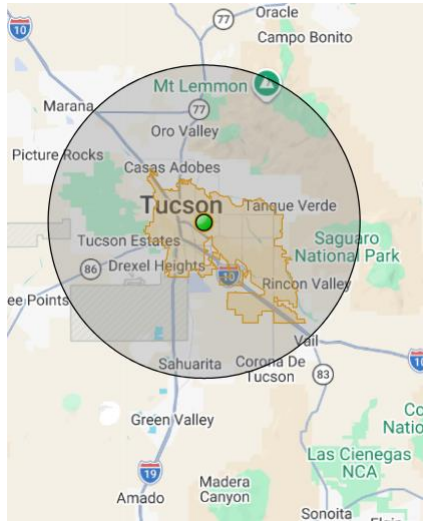
"3 miles north of Tucson"

3.0 miles?

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3.03 miles?





Convert the decimal -> fraction, Uncertainty = 1/denominator

45 degrees either direction from given cardinal direction

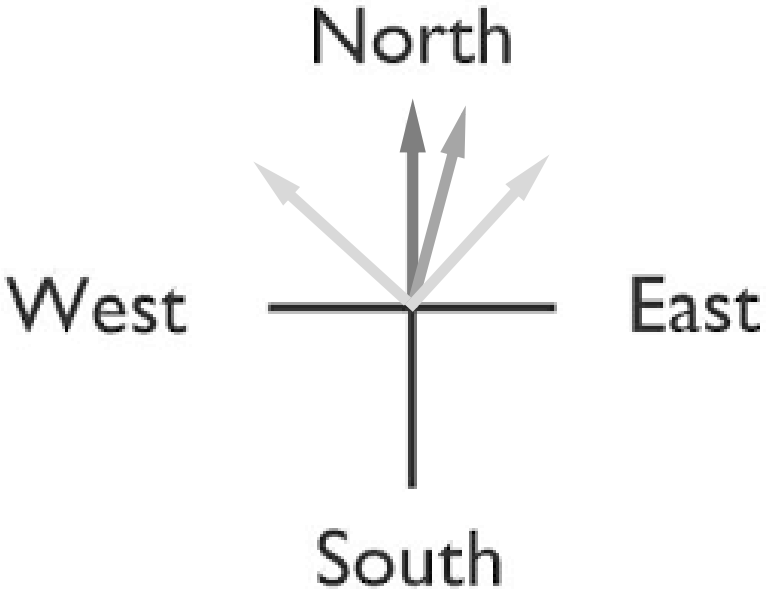
Distance from center point to furthest part of polygon

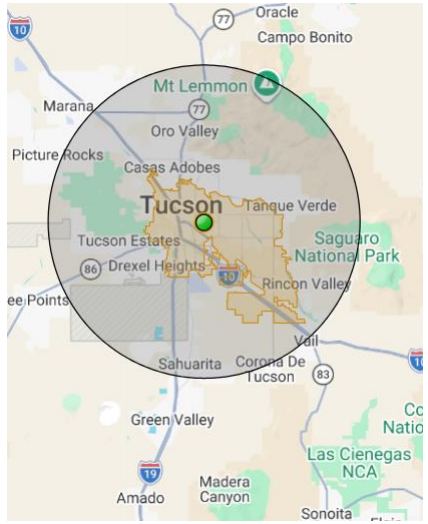
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[MaNIS/HerpNet/ORNIS Georeferencing Guidelines]

Uncertainty

Locality Description

Distance measurement precision

Direction measurement precision

Point Location

Locality extent

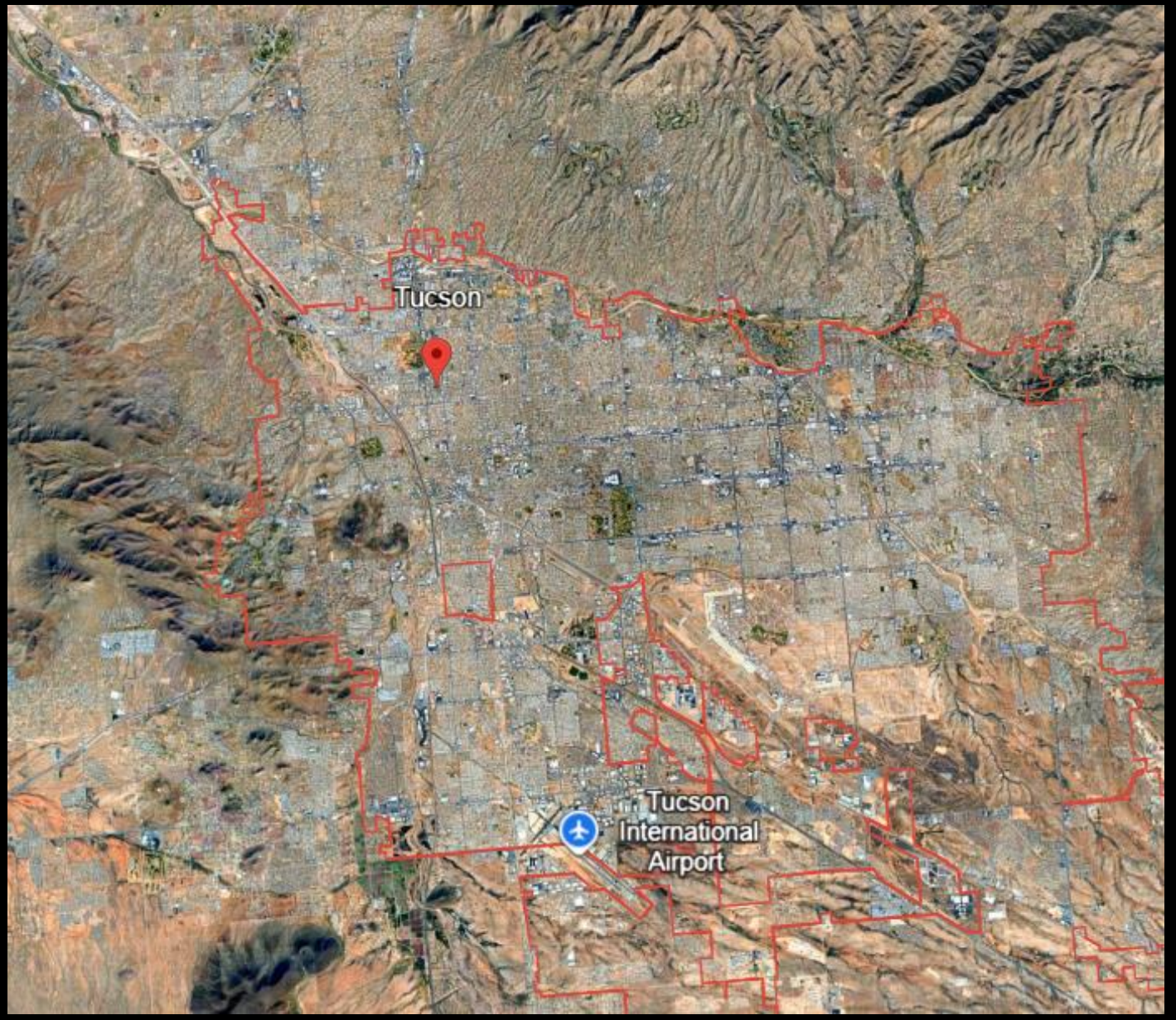
Coordinate Precision and GPS accuracy

Mapping

Map scale and Coordinate Reference Systems


Tucson

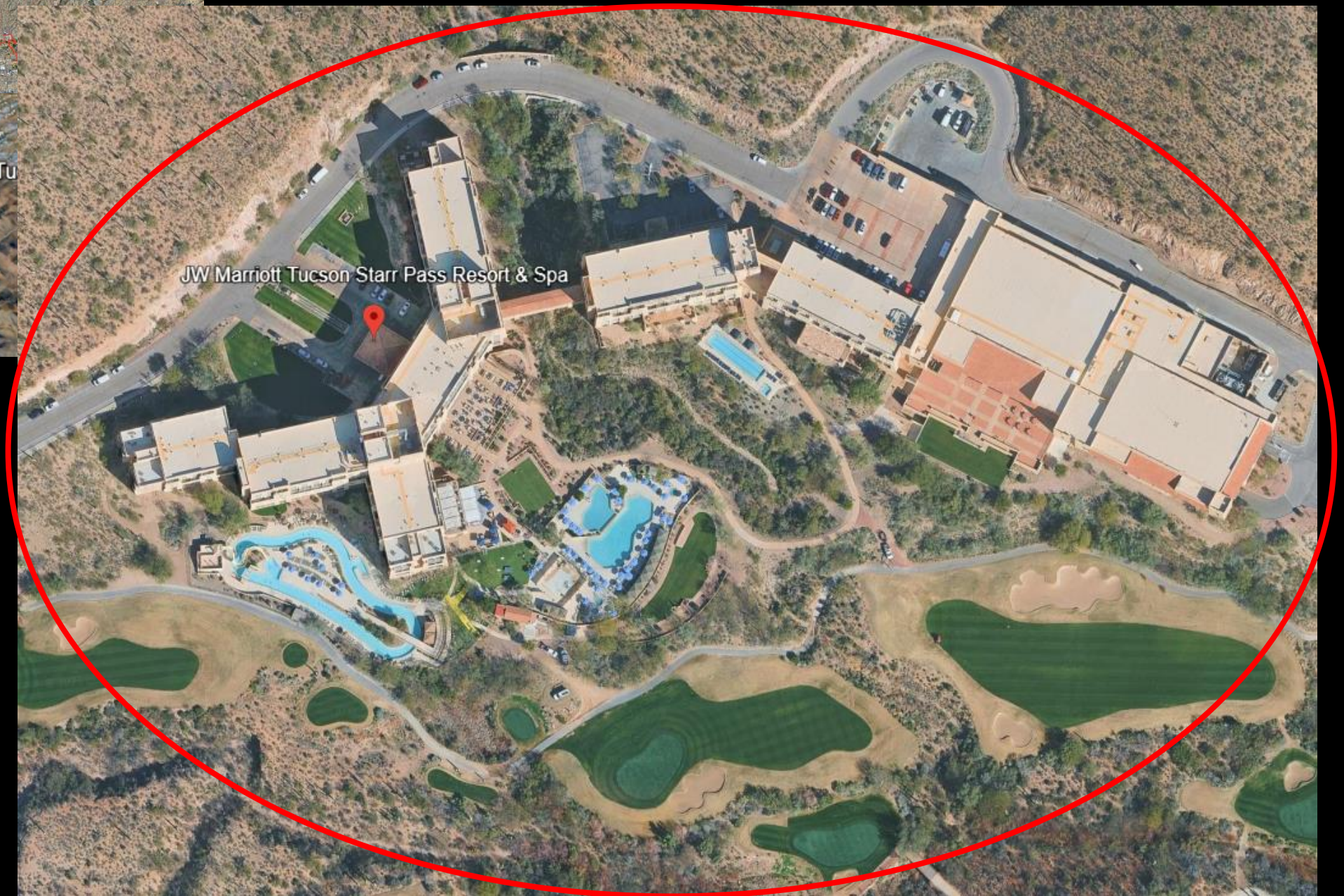

Tucson


Tucson
Tucson International Airport

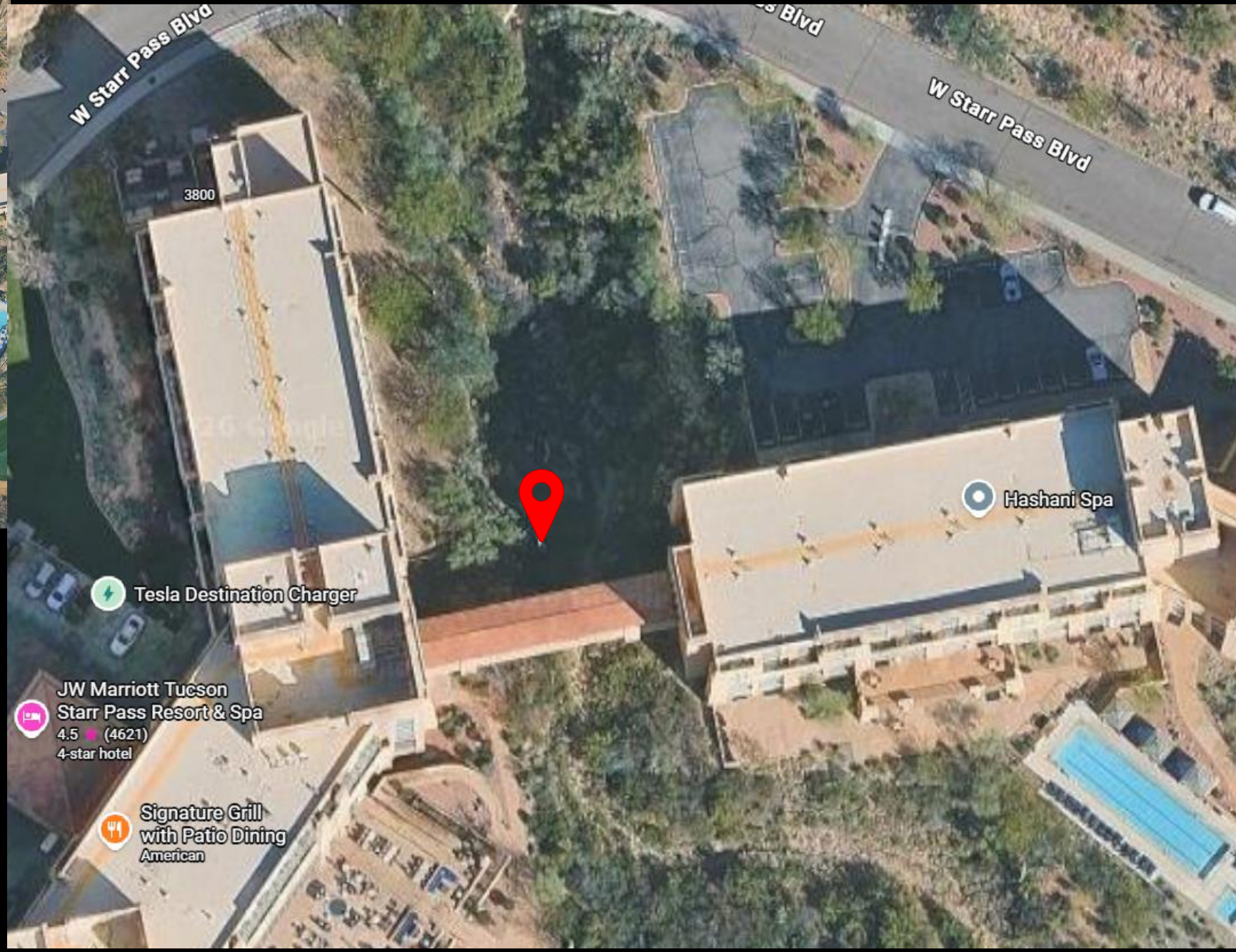

Tucson


JW Marriott Tucson Starr Pass Resort & Spa


Tucson
JW Marriott Tu


JW Marriott Tucson Starr Pass Resort & Spa

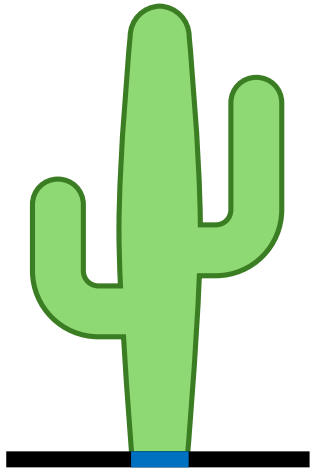

JW Marriott Tucson Starr Pass Resort & Spa


W Starr Pass Blvd
W Starr Pass Blvd
3800
Hashani Spa
Tesla Destination Charger
JW Marriott Tucson
Starr Pass Resort & Spa
4.5 (4621)
4-star hotel
Signature Grill
with Patio Dining
American

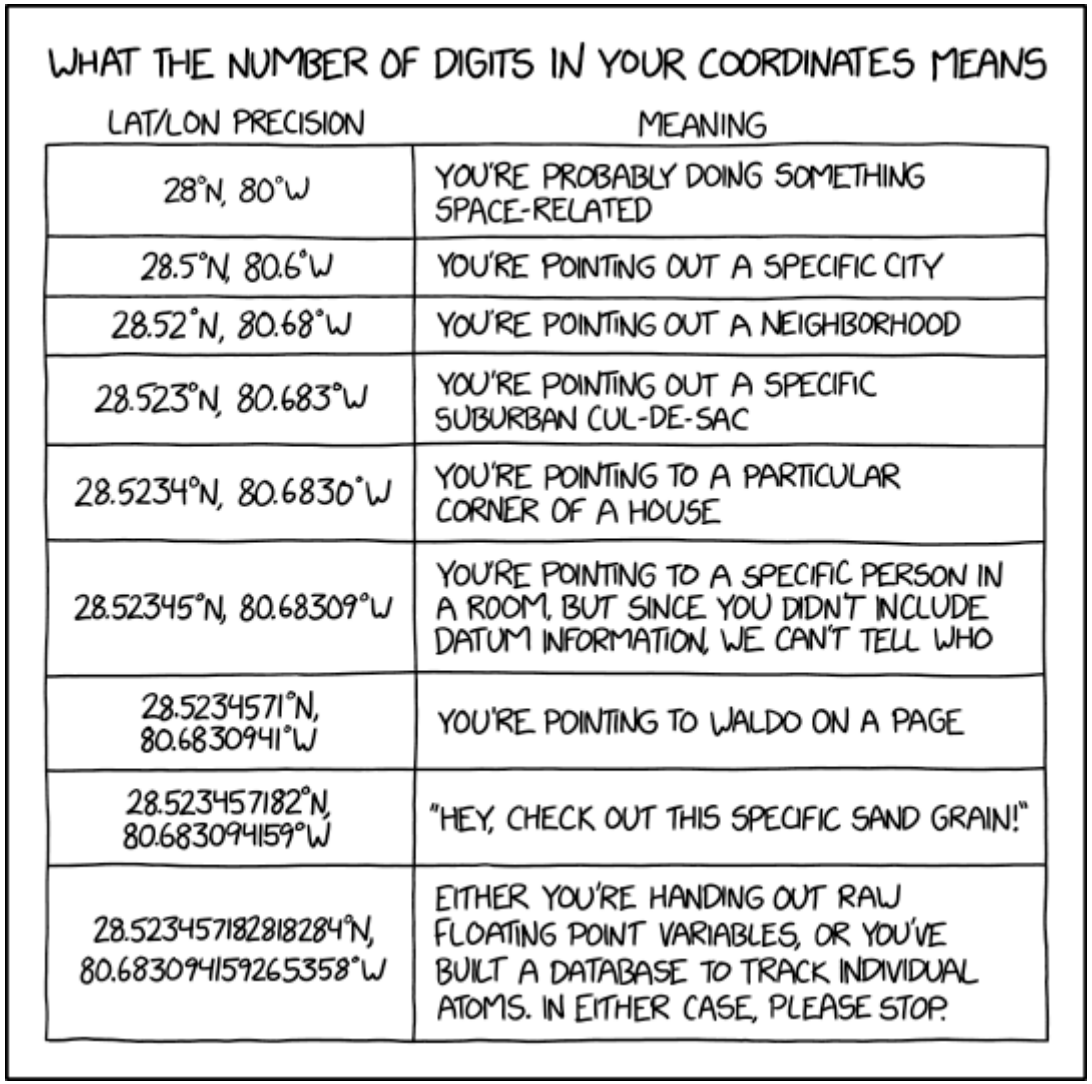

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4-star hotel
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Locality Extent

The spatial extent of a place



Coordinate Precision



WHAT THE NUMBER OF DIGITS IN YOUR COORDINATES MEANS

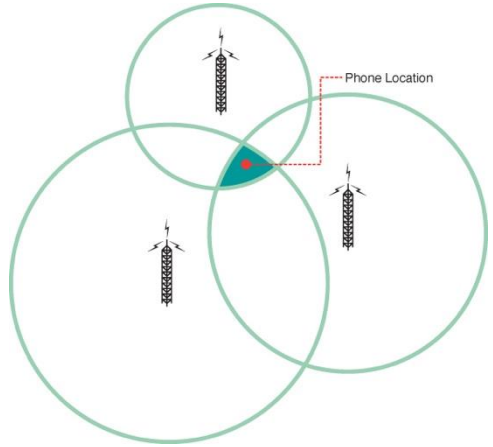
LAT/LON PRECISION	MEANING
28°N, 80°W	YOU'RE PROBABLY DOING SOMETHING SPACE-RELATED
28.5°N, 80.6°W	YOU'RE POINTING OUT A SPECIFIC CITY
28.52°N, 80.68°W	YOU'RE POINTING OUT A NEIGHBORHOOD
28.523°N, 80.683°W	YOU'RE POINTING OUT A SPECIFIC SUBURBAN CUL-DE-SAC
28.5234°N, 80.6830°W	YOU'RE POINTING TO A PARTICULAR CORNER OF A HOUSE
28.52345°N, 80.68309°W	YOU'RE POINTING TO A SPECIFIC PERSON IN A ROOM. BUT SINCE YOU DIDN'T INCLUDE DATUM INFORMATION, WE CAN'T TELL WHO
28.5234571°N, 80.6830941°W	YOU'RE POINTING TO WALDO ON A PAGE
28.523457182°N, 80.683094159°W	"HEY, CHECK OUT THIS SPECIFIC SAND GRAIN!"
28.523457182818284°N, 80.683094159265358°W	EITHER YOU'RE HANDING OUT RAW FLOATING POINT VARIABLES, OR YOU'VE BUILT A DATABASE TO TRACK INDIVIDUAL ATOMS. IN EITHER CASE, PLEASE STOP.

https://xkcd.com/2170/

Coordinate Precision

Decimal Places	Decimal Degrees	Distance	Concept
0	1.0	111 km	Can identify a Country or large Region from another
1	0.1	11.1 km	Can identify a large city from a neighbouring large city
2	0.01	1.11 km	Can identify a small village from the next
3	0.001	111 m	Can separate one neighbourhood or street from another
4	0.0001	11.1 m	Can identify an individual street or parcel of land.
5	0.00001	1.11 m	Capacity to distinguish one tree from another.
6	0.000001	11.1 cm	Measuring approximately 4 inches wide. Used for structural design & surveyance in engineering.
7	0.0000001	1.11 cm	Used for precision geographic surveying, representing the practical limit of the use of GPS.
8	0.00000001	1.11 mm	Conceptually the **width of a paper clip**. Can be used for charting volcanic movements & tectonic plate shifts.
9	0.000000001	111µm	Representative of the width of a strand of thread, now in the range of microscopy.

https://blis.com/precision-matters-critical-importance-decimal-places-five-lowest-go/



GPS accuracy





Uncertainty

Locality Description

Distance measurement precision

Direction measurement precision

Point Location

Locality extent

Coordinate Precision and GPS accuracy

Map Comparisons

Map scale and Coordinate Reference Systems

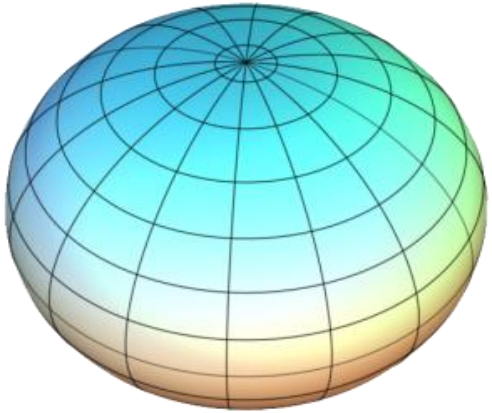
Map Comparisons



[Vox: Why all world maps are wrong]

Geodetic Datums

Model of the Earth describing the size, shape, origin, and orientation of a coordinate system for mapping Earth's surface

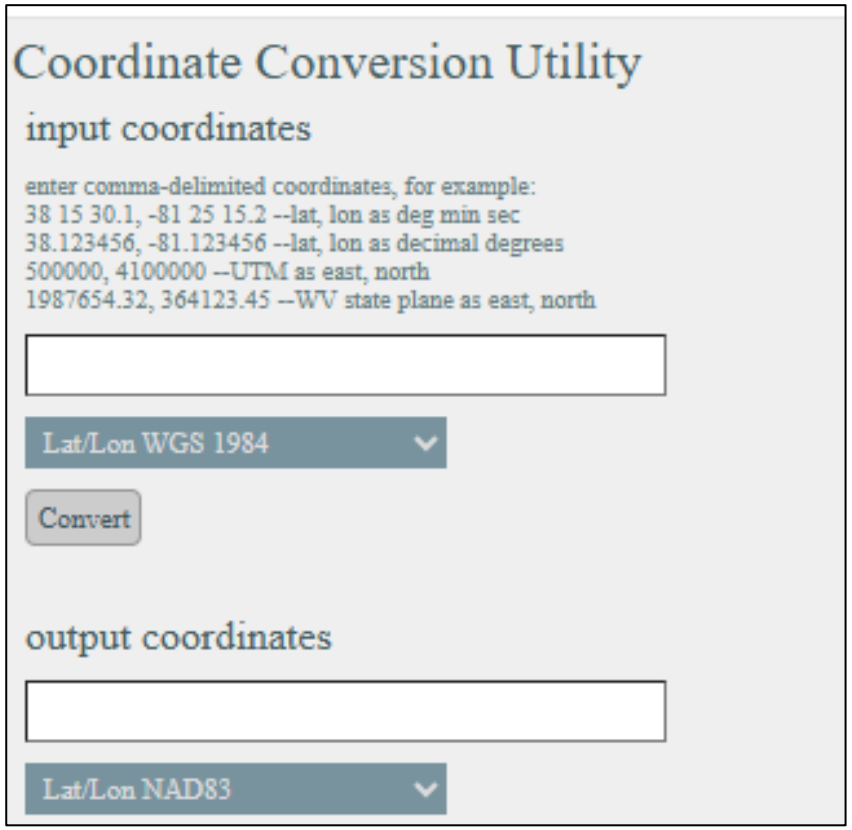


WGS84 Default for Google Maps – very common today

Others: NAD27, NAD83

Geodetic Datums

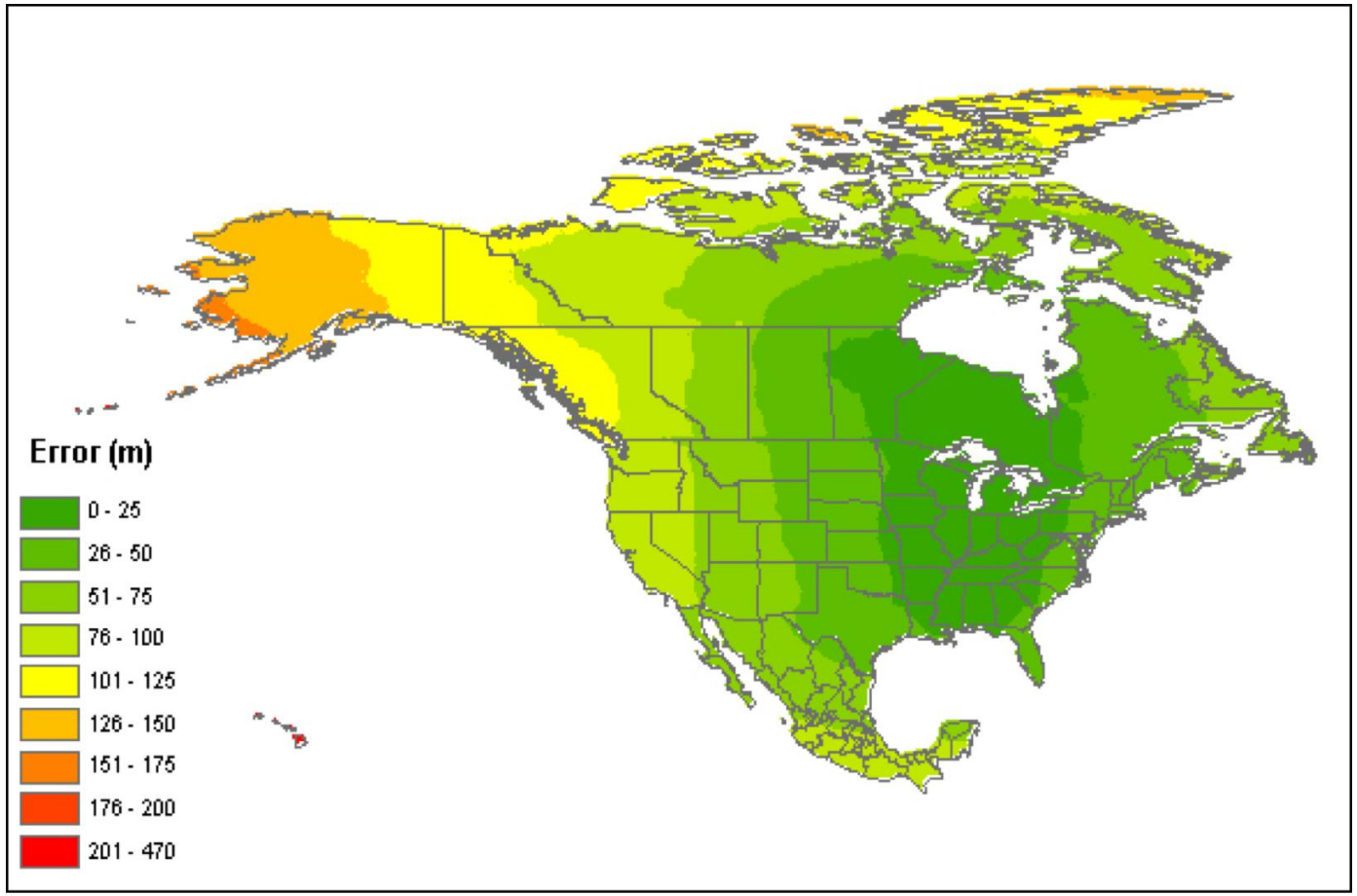
WGS84



Others:
NAD27, NAD83

https://tagis.dep.wv.gov/convert/

Error assuming NAD27 vs NAD83 or WGS84



https://georeferencing.org/docs/GeorefGuide.html

Coordinate Precision (revisited)

Uncertainty based on coordinate precision using the WGS84 reference ellipsoid

Precision	0 degrees Latitude	30 degrees Latitude	60 degrees Latitude	85 degrees Latitude
1.0 degrees	156904 m	146962 m	124605 m	112109 m
0.1 degrees	15691 m	14697 m	12461 m	11211 m
0.01 degrees	1570 m	1470 m	1247 m	1122m
0.001 degrees	157 m	147 m	125 m	113 m
0.0001 degrees	16 m	15 m	13 m	12 m
0.00001 degrees	2 m	2 m	2 m	2 m
1.0 minutes	2615 m	2450 m	2077 m	1869 m
0.1 minutes	262 m	245 m	208 m	187 m
0.01 minutes	27 m	25 m	21 m	19 m
0.001 minutes	3 m	3 m	3 m	2 m
1.0 seconds	44 m	41 m	35 m	32 m
0.1 seconds	5 m	5 m	4 m	4 m
0.01 seconds	1 m	1 m	1 m	1 m

Agenda



What is a georeference?



Why georeference?

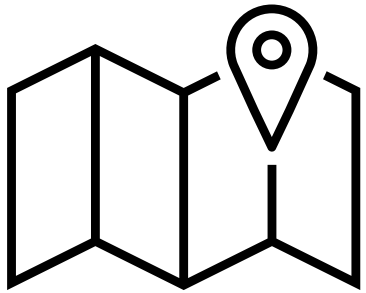


Key Concepts

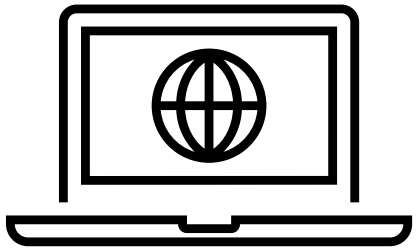


How do you georeference?

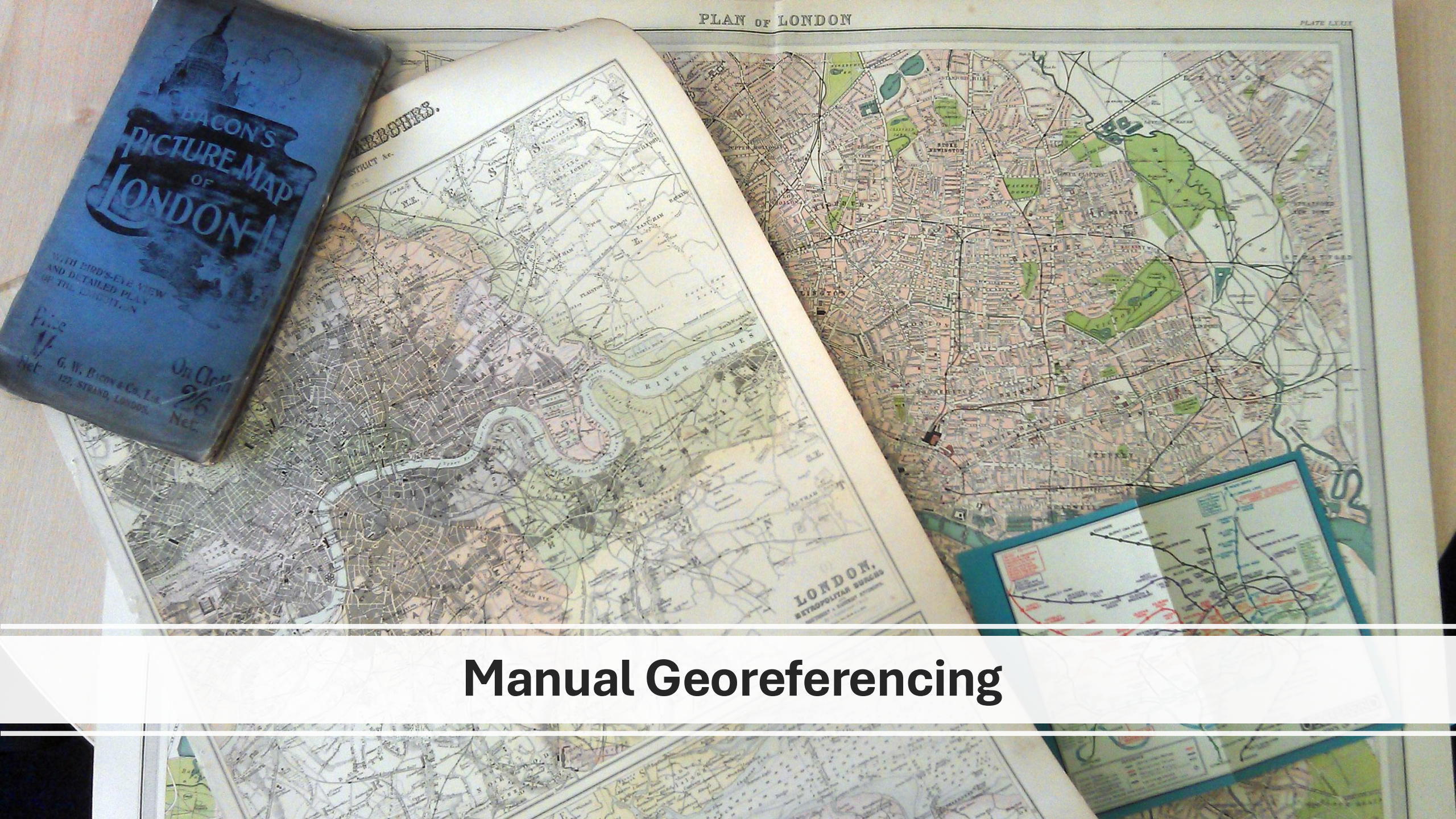
How do you georeference?



Manual Georeferencing



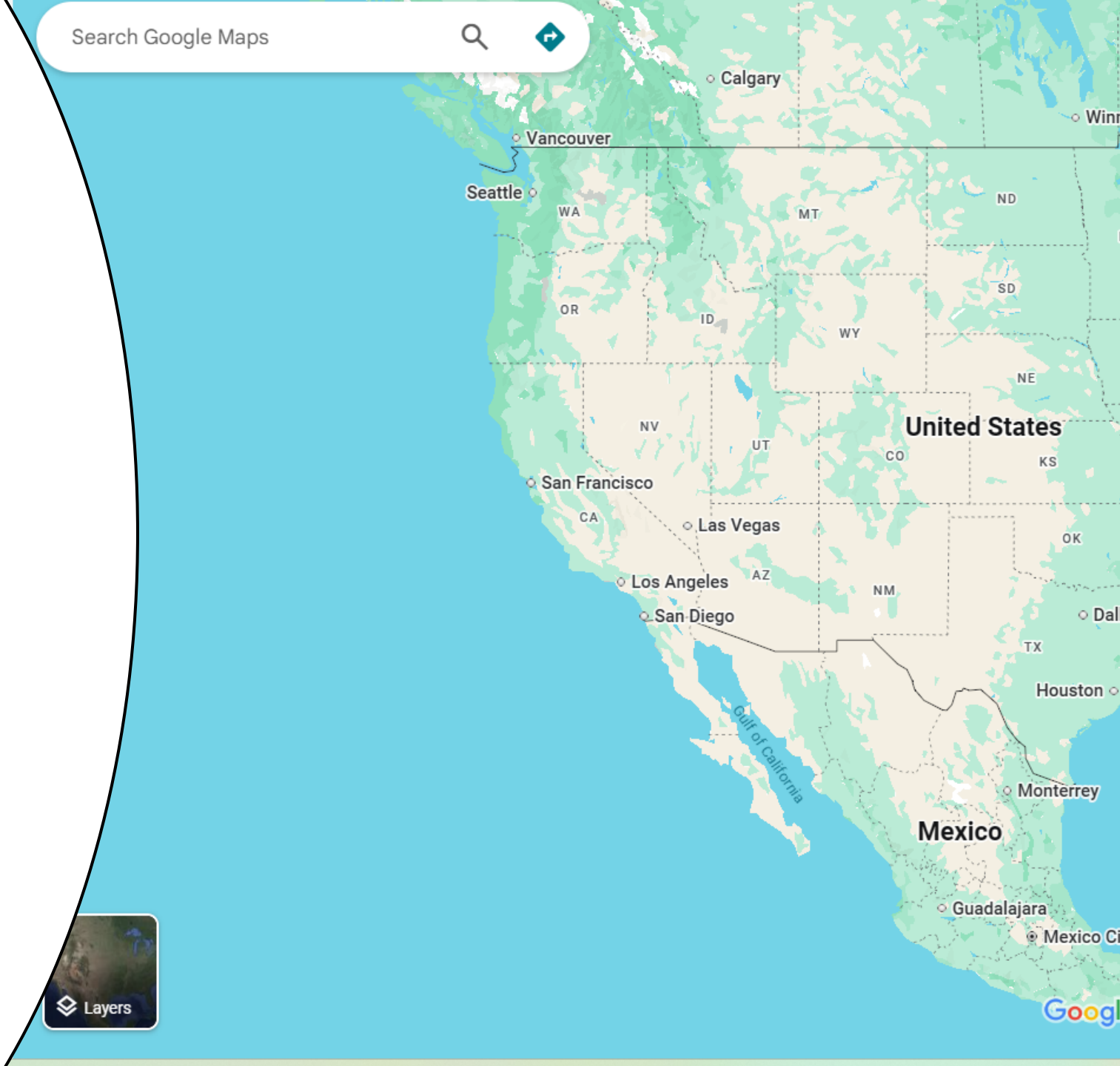
Georeferencing Calculators



Manual Georeferencing

Google Maps



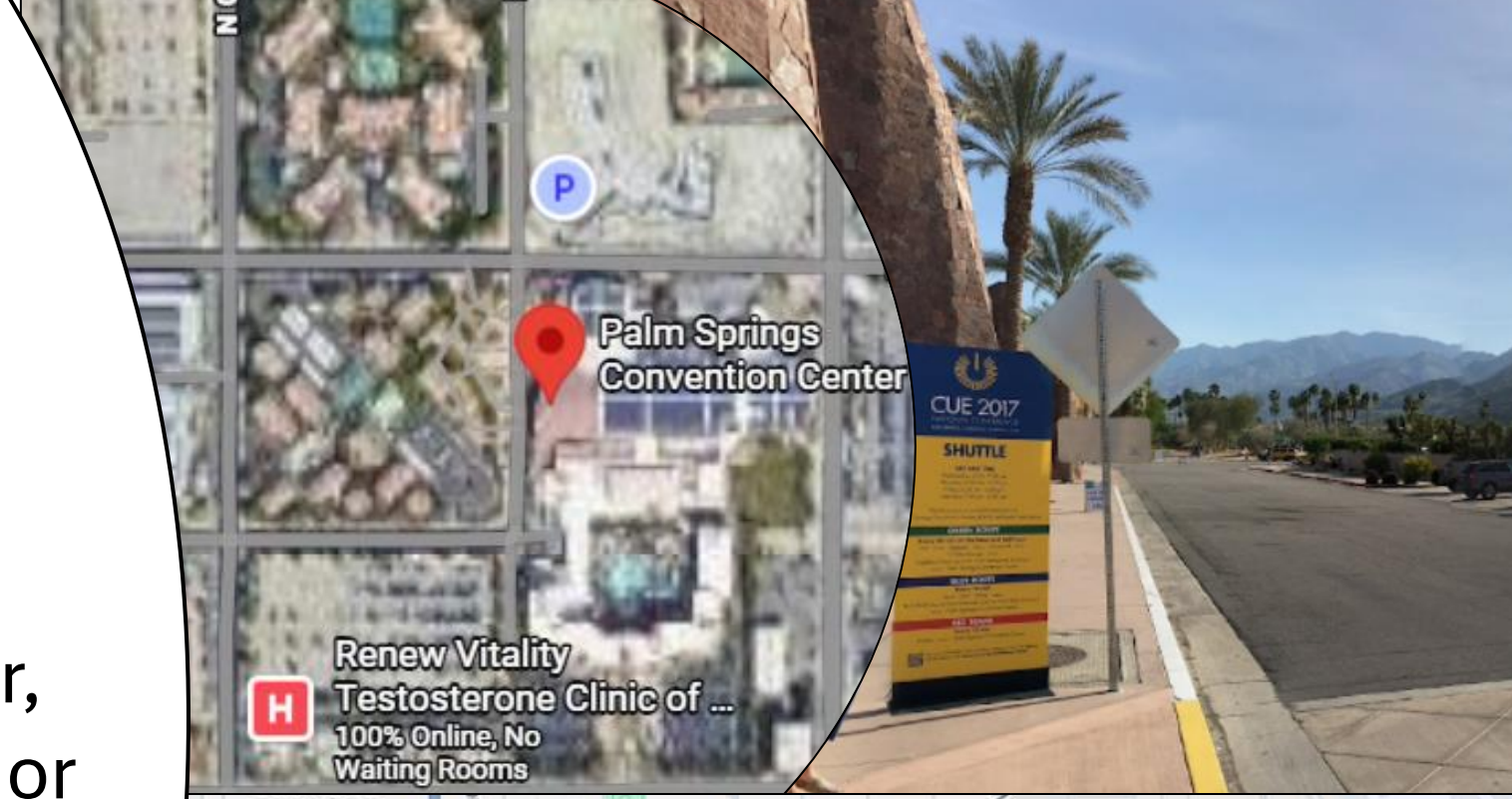

Search Google Maps
Calgary
Vancouver
Seattle
WA
MT
ND
OR
ID
SD
WY
NE
NV
UT
CO
KS
United States
San Francisco
CA
Las Vegas
OK
AZ
NM
Los Angeles
San Diego
TX
Houston
Gulf of California
Monterrey
Mexico
Guadalajara
Layers

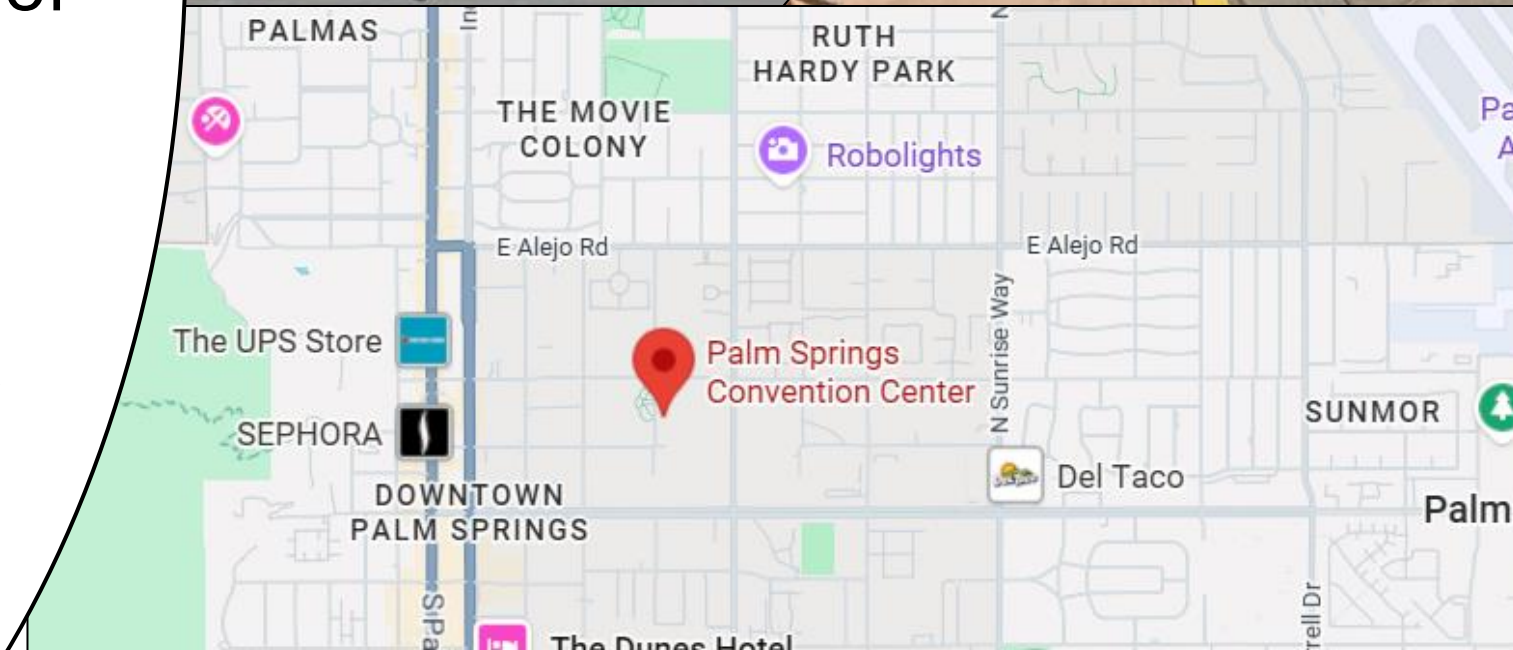
Google Maps

Easily searchable

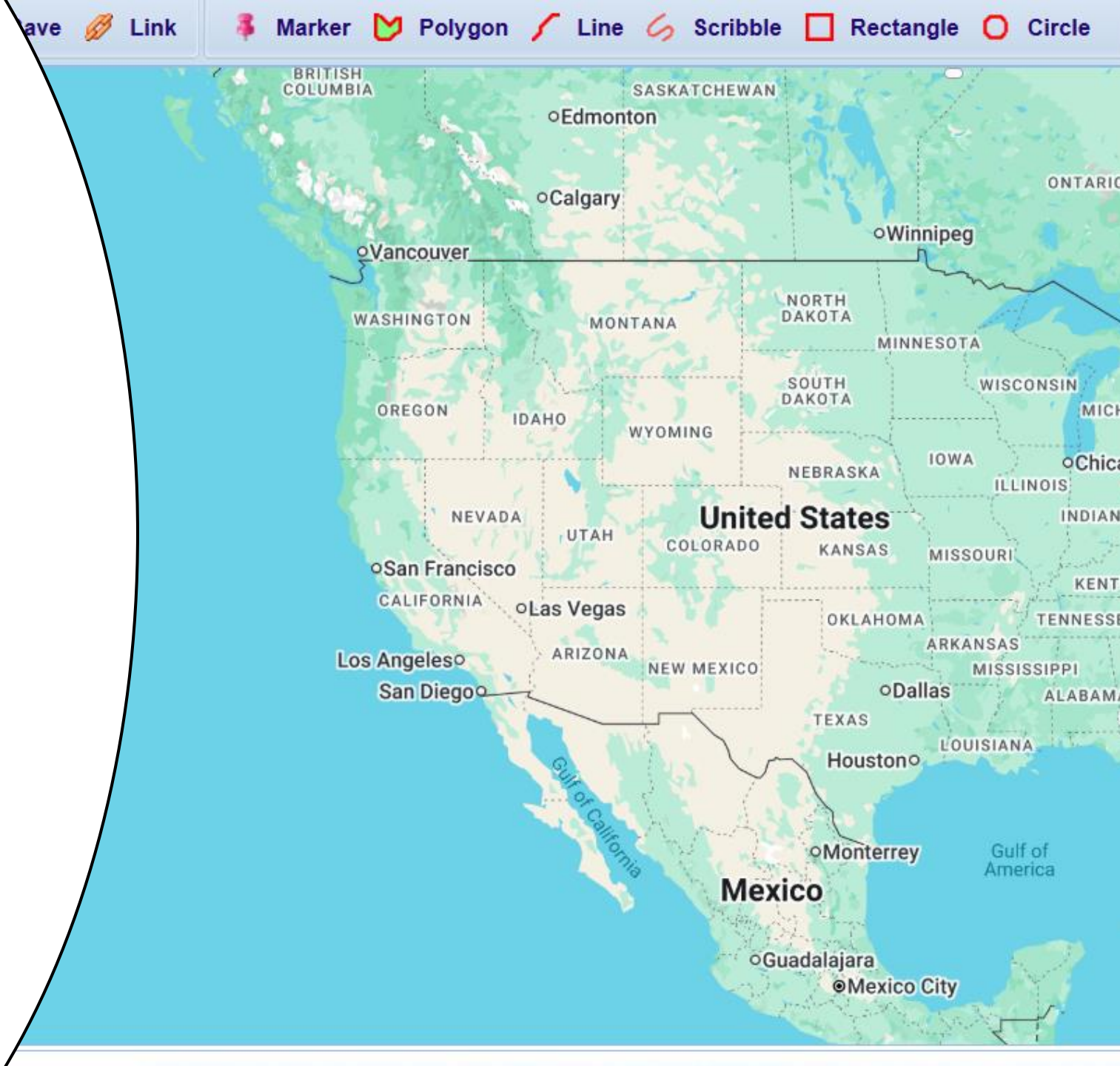
Directions for traveling by car, bike, public transportation, or foot

Map views: map, satellite, terrain, Google Street View





GmapGIS

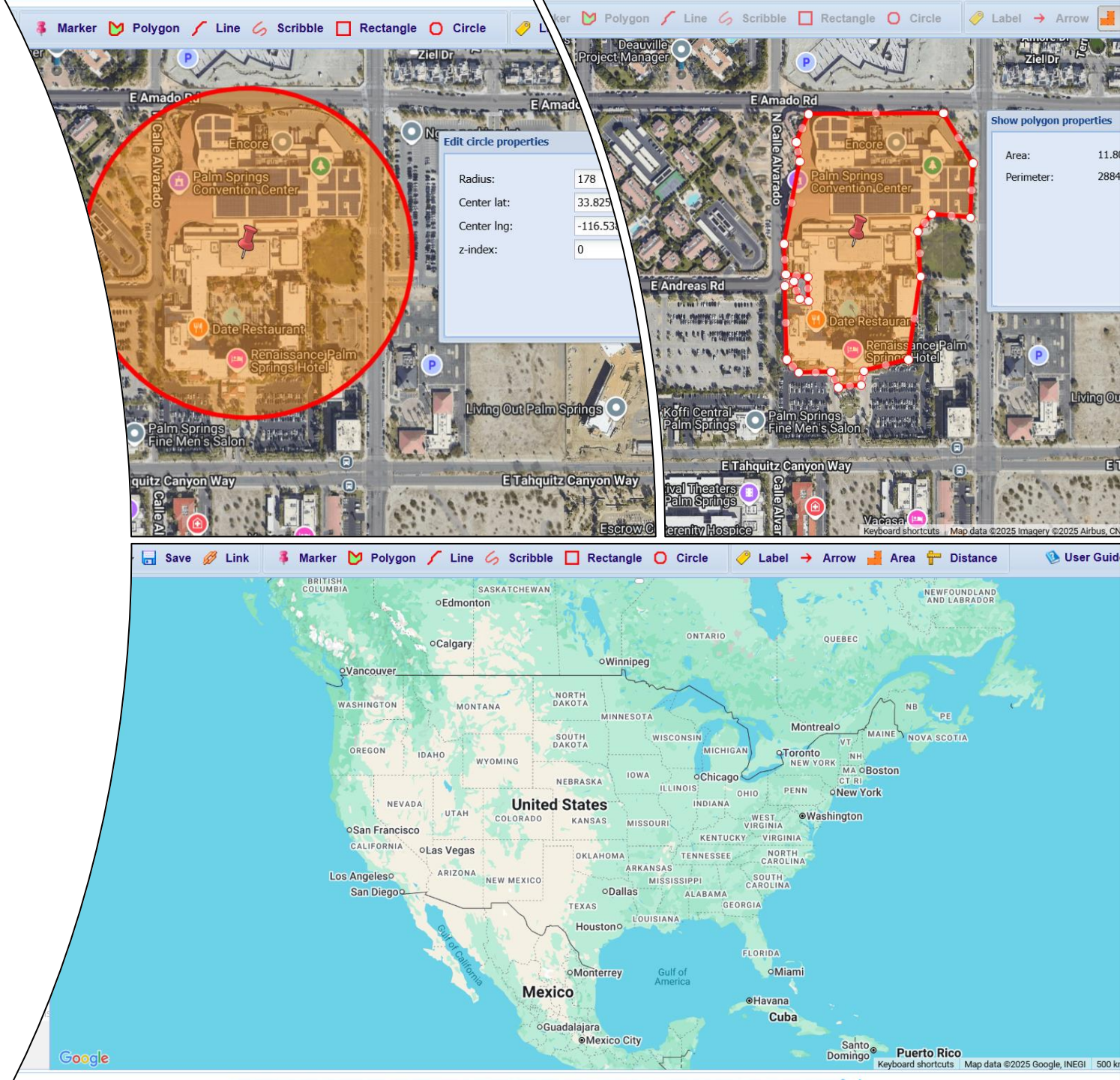

ave
Link
Marker
Polygon
Line
Scribble
Rectangle
Circle
BRITISH COLUMBIA
SASKATCHEWAN
Edmonton
Calgary
Winnipeg
Vancouver
WASHINGTON
MONTANA
NORTH DAKOTA
MINNESOTA
OREGON
IDAHO
WYOMING
SOUTH DAKOTA
WISCONSIN
NEBRASKA
IOWA
ILLINOIS
NEVADA
UTAH
COLORADO
KANSAS
MISSOURI
United States
San Francisco
CALIFORNIA
Las Vegas
OKLAHOMA
ARKANSAS
ARIZONA
NEW MEXICO
MISSISSIPPI
Los Angeles
San Diego
Dallas
TEXAS
LOUISIANA
Houston
Gulf of California
Monterrey
Gulf of America
Mexico
Guadalajara
Mexico City

GmapGIS

Google Map Layers

Zoom to Coordinates

Easy to draw/measure uncertainty



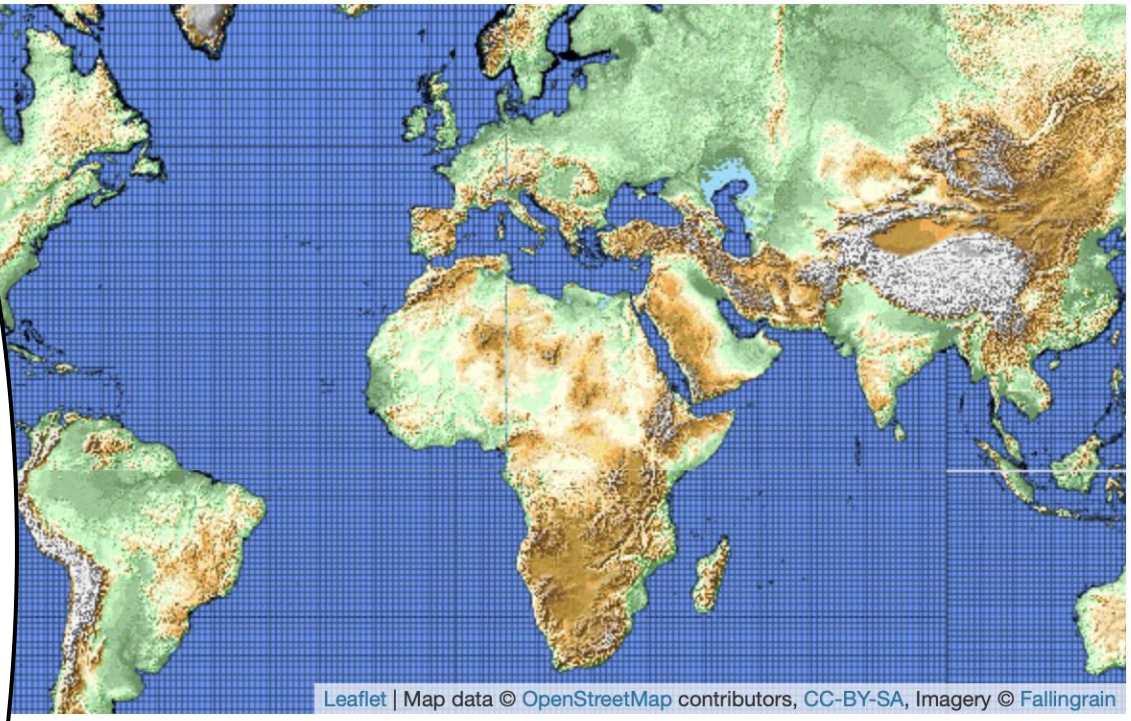
Falling Rain

https://www.fallingrain.com/world/

Worldwide gazetteer for cities and towns

Great for **hard-to-find localities, especially outside US**

l Gazetteer Version 2.3



Cartier Islands

a
man Islands
lgaria
urundi

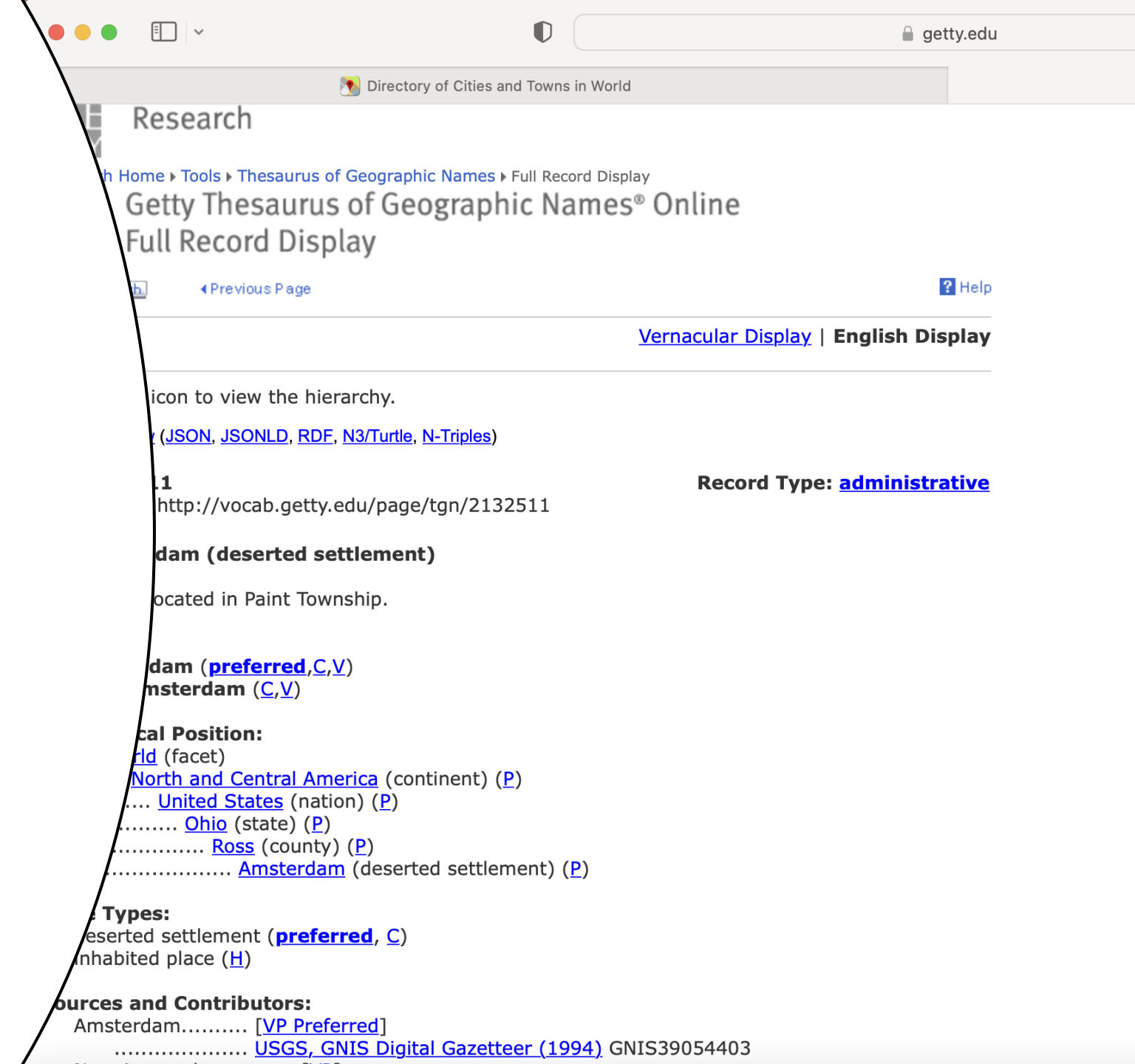
AC Antigua & Barbuda
AG Algeria
AM Armenia
AR Argentina
AU Austria
AY Antarctica
BC Botswana
BF Bahamas, The
BK Bosnia & Herzegovina
BN Benin
BR Brazil
BV Bouvet Island
CA Canada

Getty Thesaurus of Geographic Names (TGN)

http://bit.ly/Getty-TGN

Useful for **alternative and old names**

Degrees-minutes, not coordinates



Georeferencing Calculators



GEOLocate

1 **Data Entry & Preparation**

2 **Automated Processing**

3 **Manual Verification**

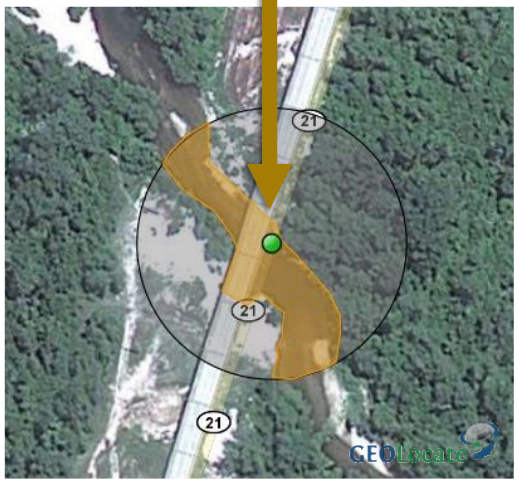
pushepatapa creek, trib. to pearl river, 7.8 miles north of bogalusa at hwy 21; Washington; LA; USA

↓

Georeferencing Algorithm

↓

Visualize, verify & adjust output coordinates & uncertainties



latitude: **30.88797**
longitude: **-89.83601**

uncertainty radius: **48m**

uncertainty polygon: **30.88823,-89.83641, 30.88815,-89.83634, 30.88808,-89.83622…**

Single Reference Workflow

Home | Web Application | Collaborative Georeferencing | Developer Resources | Education & Outreach | Support and Contacts

Web Based Clients

The following based web clients are available to allow you to georeference data directly from your web browser:



Standard Client

Simply type in your locality description and get back georeferenced results. Start here if you are new to GEOLocate.



Batch (File Based) Client

Allows you to upload a .csv file and batch process it. (file formatting instructions)



Collaborative Georeferencing Client

Utilizes the collaborative georeferencing framework. Ideal for largescale multi-institution projects. (https link)

Note: if you use the secure SSL (HTTPS) link, please make sure you browser is configured to allow mixed mixed content, or you may see a blank map. Here are SSL configuration instructions for various browsers: in English and in Spanish (special thanks to David Draper for the Spanish translation).

Embeddable client

- A streamlined web client for the purpose of embedding in other web applications.
  Sample link demonstrating use this client.
  Documentation link on how to craft URLs for this client.

Other Clients:

- Arctos
- Specify
- Symbiota
- Tropicos

Know of any other web based clients using GEOLocate? Let us know and we will be happy to list them.

GEOLocate Web Application



Workbench | Results

Georeference | Options | Draw polygon | Place marker | Measure

Locality String: Palm Springs Conference Center

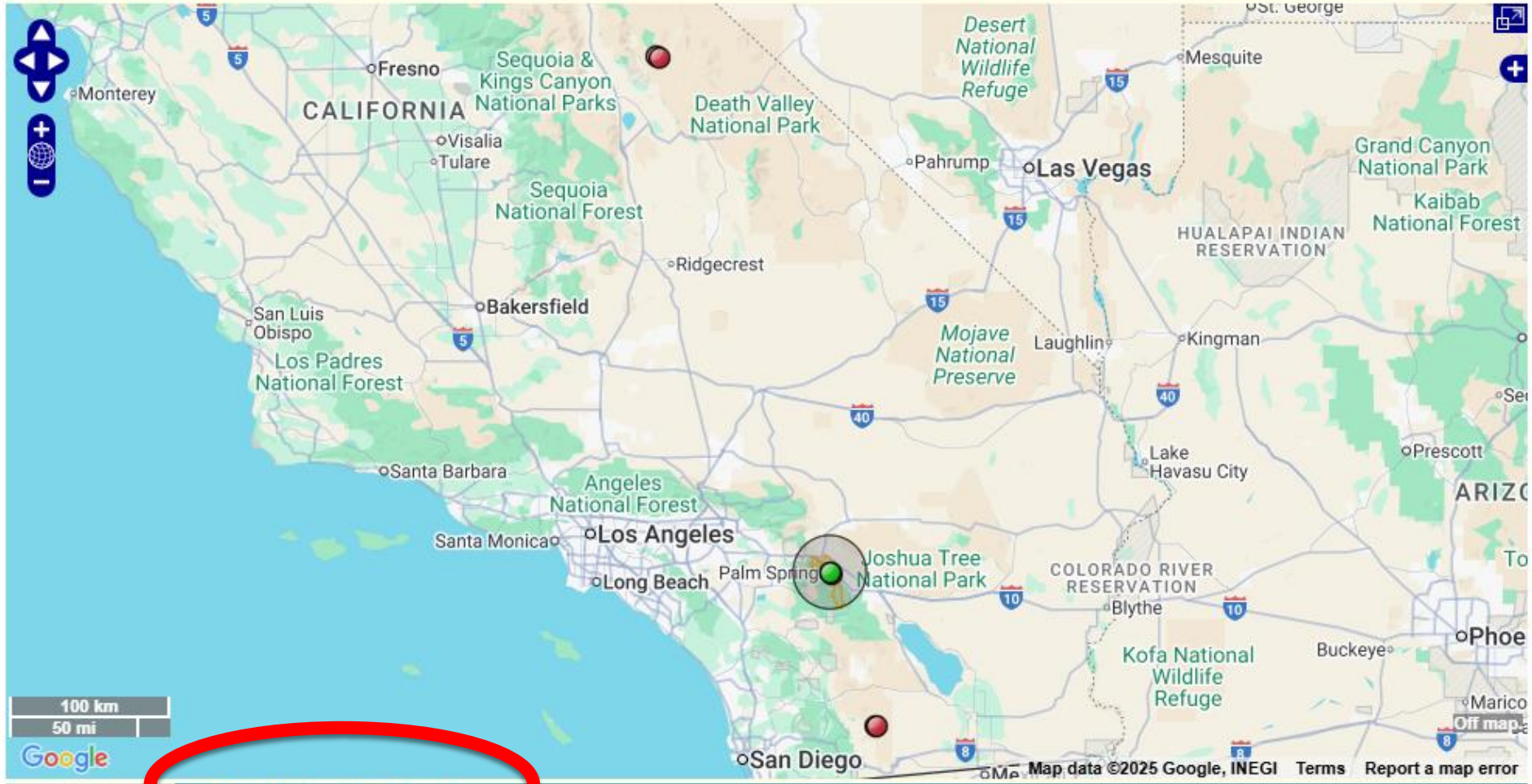
Country: United States of America

State: California

County:

latitude: longitude: uncertainty: polygon

GEOLocate Web Application



Workbench

6 possible locations found

Georeference | Options | Clear Polygon | Draw polygon | Place marker | Measure

Locality String: Palm Springs Conference Center

Country: United States of America

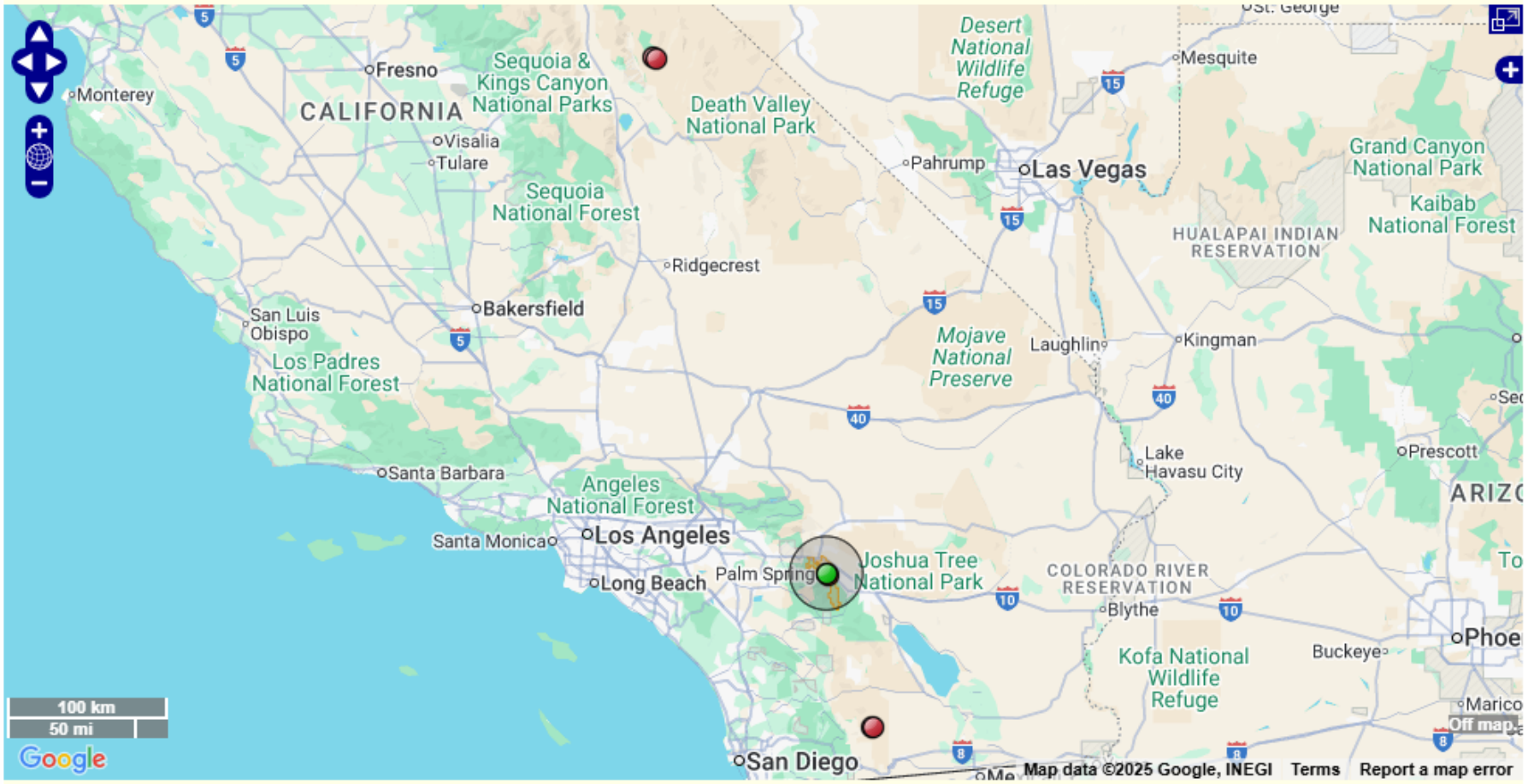
State: California

County:

latitude: 33.830296 longitude: -116.545292 uncertainty: 24010 m polygon

```
33.830296       -116.545292     24010
33.8880160149,-116.675348944,33.8879420149,-116.671143944,33.8879410149,
-116.671060944,33.8878960149,-116.667112944,33.8918310149,-116.667088944
```

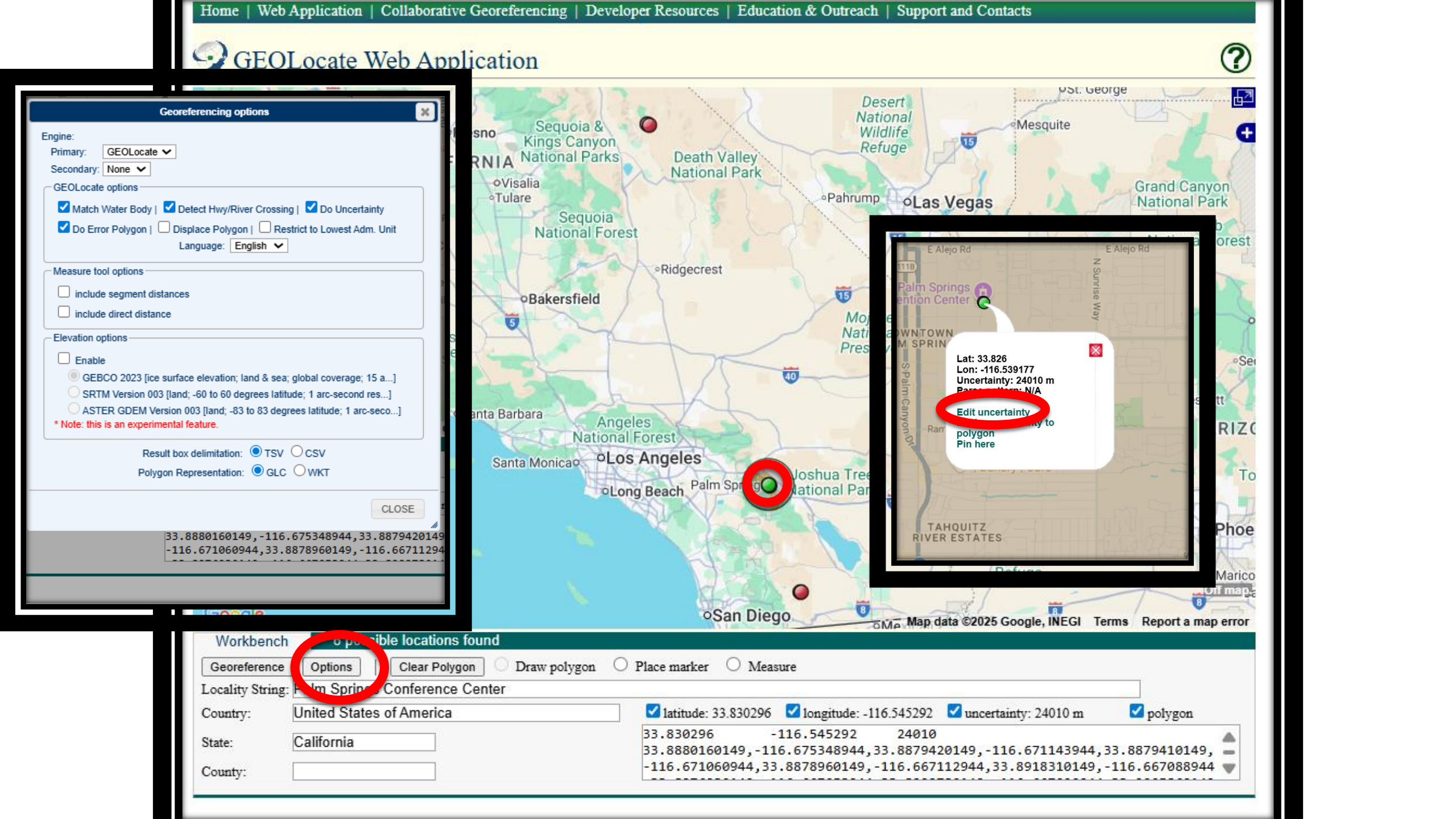
GEOLocate Web Application



Workbench | 6 possible locations found

- lat: 33.830296, lon: -116.545292, pattern: PALM SPRINGS, error polygon: 33.8880160149,-116.675348944..., uncertainty: 24010 m, precision: Low (44)
- lat: 33.8167, lon: -116.5333, pattern: PALM SPRINGS, error polygon: Unavailable, uncertainty: Unavailable, precision: Low (42)
- lat: 36.813057, lon: -117.766016, pattern: PALM SPRING, error polygon: Unavailable, uncertainty: Unavailable, precision: Low (28)
- lat: 32.919991, lon: -116.218074, pattern: PALM SPRING, error polygon: Unavailable, uncertainty: Unavailable, precision: Low (28)
- lat: 32.9167, lon: -116.2167, pattern: PALM SPRING, error polygon: Unavailable, uncertainty: Unavailable, precision: Low (28)
- lat: 36.8, lon: -117.75, pattern: PALM SPRING, error polygon: Unavailable, uncertainty: Unavailable, precision: Low (28)

Remove Secondary Points


Home | Web Application | Collaborative Georeferencing | Developer Resources | Education & Outreach | Support and Contacts
GEOLocate Web Application
Georeferencing options
Engine:
Primary: GEOLocate
Secondary: None
GEOLocate options
Match Water Body | Detect Hwy/River Crossing | Do Uncertainty
Do Error Polygon | Displace Polygon | Restrict to Lowest Adm. Unit
Language: English
Measure tool options
include segment distances
include direct distance
Elevation options
Enable
GEBCO 2023 [ice surface elevation; land & sea; global coverage; 15 a...]
SRTM Version 003 [land; -60 to 60 degrees latitude; 1 arc-second res...]
ASTER GDEM Version 003 [land; -83 to 83 degrees latitude; 1 arc-seco...]
* Note: this is an experimental feature.
Result box delimitation: TSV CSV
Polygon Representation: GLC WKT
CLOSE
33.8880160149,-116.675348944,33.8879420149
-116.671060944,33.8878960149,-116.66711294
Desert National Wildlife Refuge
Mesquite
Sequoia & Kings Canyon National Parks
Death Valley National Park
Visalia
Tulare
Pahrump
Las Vegas
Grand Canyon National Park
Sequoia National Forest
Ridgecrest
Bakersfield
Angeles National Forest
Santa Barbara
Santa Monica
Los Angeles
Long Beach
Palm Springs
Joshua Tree National Park
San Diego
E Alejo Rd
N Sunrise Way
S Palm Canyon Dr
Palm Springs Convention Center
TAHQUITZ RIVER ESTATES
Lat: 33.826
Lon: -116.539177
Uncertainty: 24010 m
Edit uncertainty
polygon
Pin here
Map data ©2025 Google, INEGI Terms Report a map error
Workbench
8 possible locations found
Georeference
Options
Clear Polygon
Draw polygon
Place marker
Measure
Locality String: Palm Springs Convention Center
Country: United States of America
State: California
County:
latitude: 33.830296
longitude: -116.545292
uncertainty: 24010 m
polygon
33.830296 -116.545292 24010
33.8880160149,-116.675348944,33.8879420149,-116.671143944,33.8879410149,
-116.671060944,33.8878960149,-116.667112944,33.8918310149,-116.667088944

GEOLocate Web Application



Workbench | 6 possible locations found

Georeference | Options | Clear Polygon | Draw polygon | Place marker | Measure

Locality String: Palm Springs Conference Center

Country: United States of America

State: United States of America

County:

latitude: 33.824966 longitude: -116.538254 uncertainty: 218 m polygon

```
33.824966       -116.538254     218
33.8880160149,-116.675348944,33.8879420149,-116.671143944,33.8879410149,
-116.671060944,33.8878960149,-116.667112944,33.8918310149,-116.667088944
```

Home | Web Application | Collaborative Georeferencing | Developer Resources | Education & Outreach | Support and Contacts

GEOLocate Web Application



Workbench | 6 possible locations found

Georeference | Options | Clear Polygon | Draw polygon | Place marker | Measure

Locality String: Palm Springs Conference Center

Country: United States of America

State: United States of America

County:

latitude: 33.824966 | longitude: -116.53825 | uncertainty: 218 m | polygon

33.824966 -116.538254 218

33.8880160149,-116.675348944,33.8879426[illegible]43944,33.8879410149,
-116.671060944,33.8878960149,-116.667112944,33.8918310149,-116.667088944





Batch Workflow

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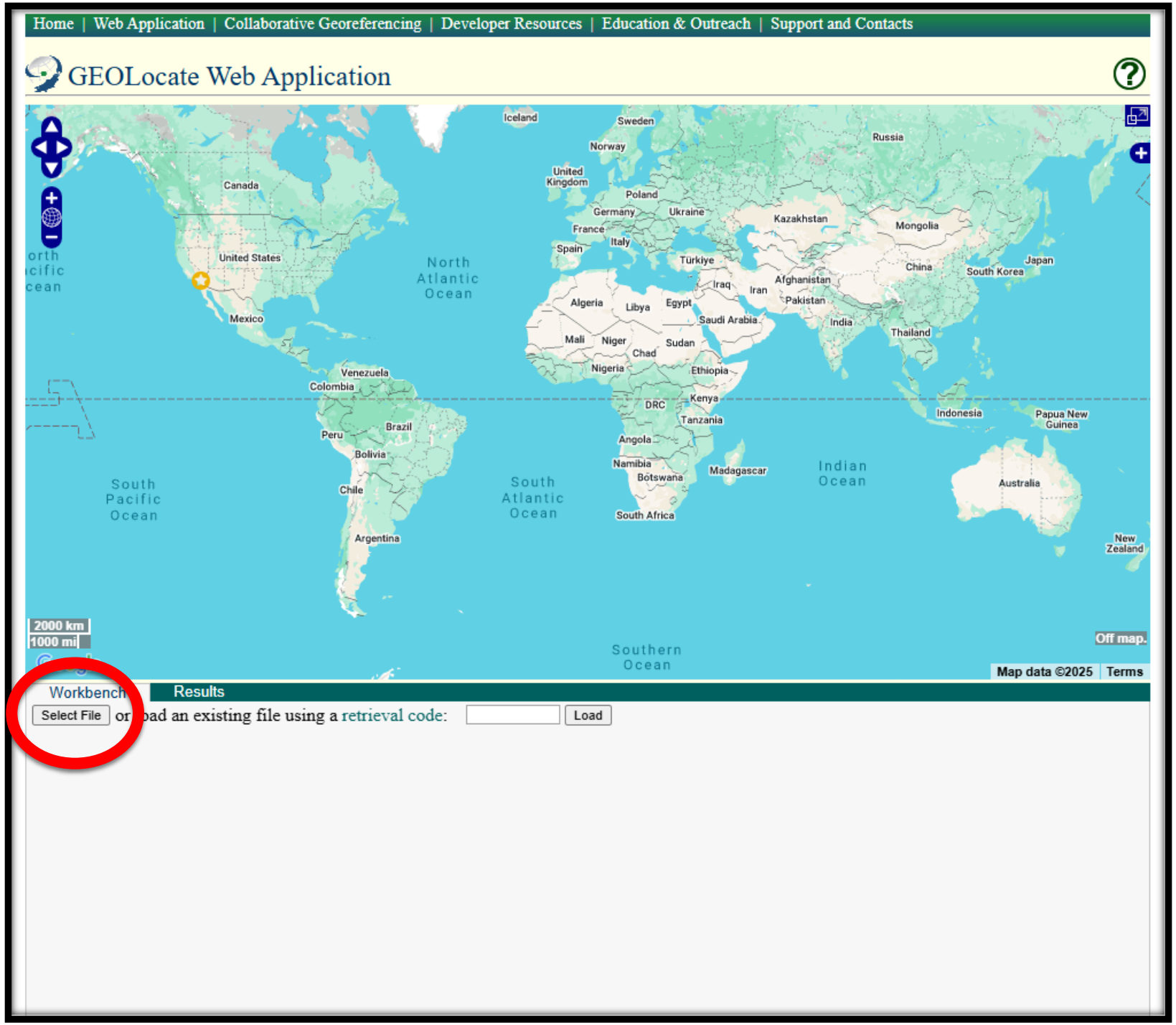
Embeddable client

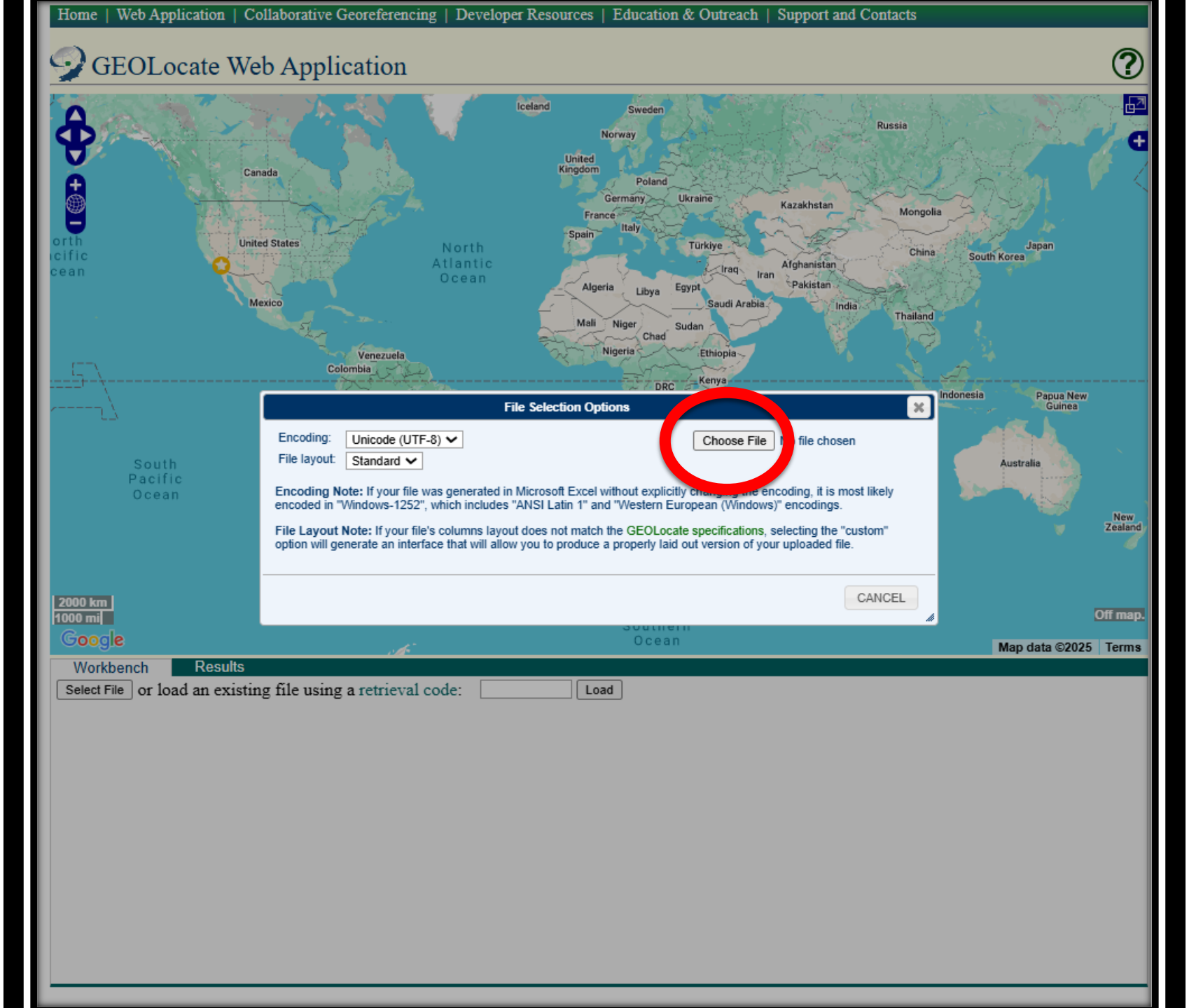
- A streamlined web client for the purpose of embedding in other web applications.
  Sample link demonstrating use this client.
  Documentation link on how to craft URLs for this client.

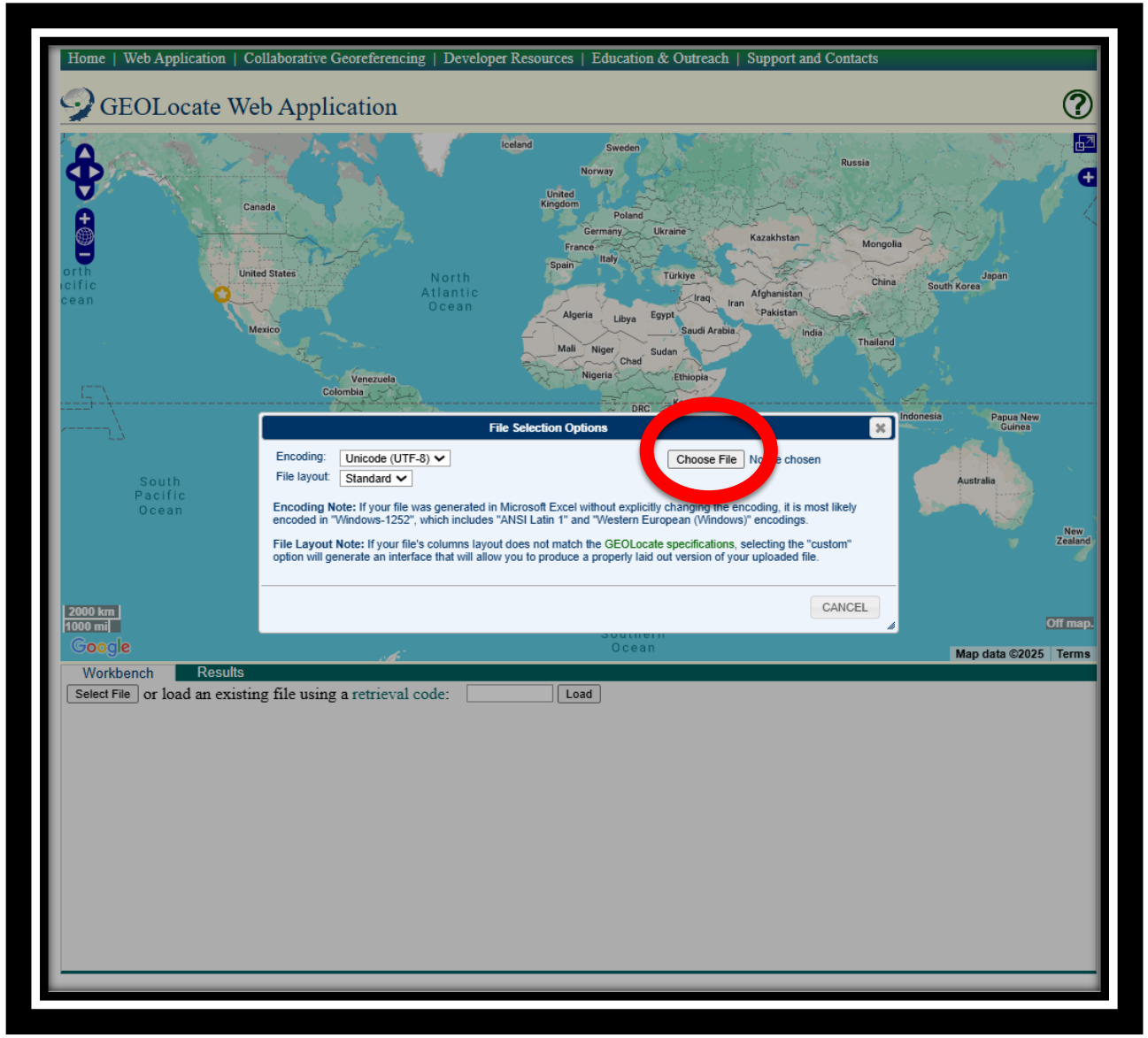
Other Clients:

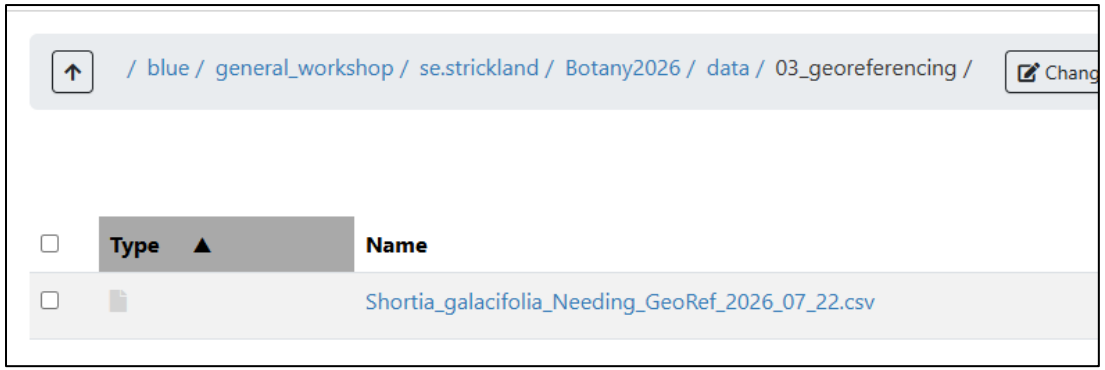
- Arctos
- Specify
- Symbiota
- Tropicos

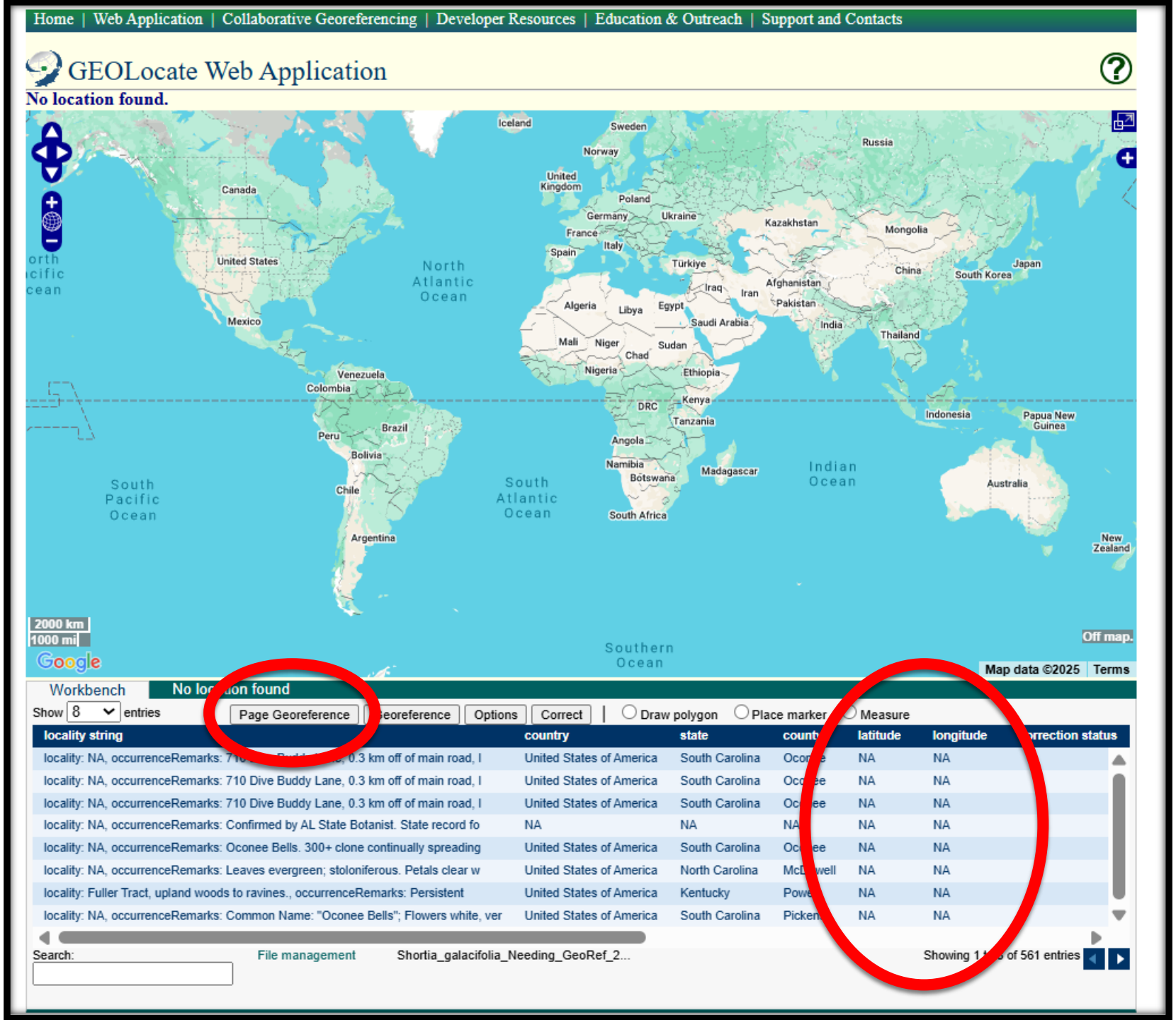
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Home | Web Application | Collaborative Georeferencing | Developer Resources | Education & Outreach | Support and Contacts
GEOLocate Web Application
Iceland
Sweden
Norway
Russia
United Kingdom
Canada
Poland
Germany
Ukraine
Kazakhstan
Mongolia
France
Spain
Italy
Türkiye
China
Japan
South Korea
United States
North Atlantic Ocean
Afghanistan
Iraq
Iran
Pakistan
Algeria
Libya
Egypt
Saudi Arabia
India
Thailand
Mexico
Mali
Niger
Chad
Sudan
Nigeria
Ethiopia
Venezuela
Colombia
Kenya
DRC
Tanzania
Indonesia
Papua New Guinea
Peru
Brazil
Angola
Bolivia
Namibia
Botswana
Madagascar
Indian Ocean
South Pacific Ocean
South Atlantic Ocean
Chile
Australia
South Africa
Argentina
New Zealand
2000 km
1000 mi
Off map.
Southern Ocean
Map data ©2025
Terms
Workbench
Results
Select File
or load an existing file using a retrieval code:
Load


Home | Web Application | Collaborative Georeferencing | Developer Resources | Education & Outreach | Support and Contacts
GEOLocate Web Application
Iceland
Sweden
Norway
Russia
United Kingdom
Canada
Poland
Germany
Ukraine
Kazakhstan
Mongolia
France
Italy
Spain
United States
North Atlantic Ocean
Türkiye
China
South Korea
Japan
Iraq
Afghanistan
Iran
Pakistan
Algeria
Libya
Egypt
Saudi Arabia
India
Thailand
Mexico
Mali
Niger
Sudan
Chad
Nigeria
Ethiopia
Venezuela
Colombia
DRC
Kenya
Indonesia
Papua New Guinea
Australia
New Zealand
South Pacific Ocean
File Selection Options
Encoding:
Unicode (UTF-8)
File layout:
Standard
Choose File
file chosen
Encoding Note: If your file was generated in Microsoft Excel without explicitly changing the encoding, it is most likely encoded in "Windows-1252", which includes "ANSI Latin 1" and "Western European (Windows)" encodings.
File Layout Note: If your file's columns layout does not match the GEOLocate specifications, selecting the "custom" option will generate an interface that will allow you to produce a properly laid out version of your uploaded file.
CANCEL
2000 km
1000 mi
Google
Off map.
Map data ©2025
Terms
Workbench
Results
Select File
or load an existing file using a retrieval code:
Load


Home | Web Application | Collaborative Georeferencing | Developer Resources | Education & Outreach | Support and Contacts
GEOLocate Web Application
File Selection Options
Encoding: Unicode (UTF-8)
File layout: Standard
Choose File
Encoding Note: If your file was generated in Microsoft Excel without explicitly changing the encoding, it is most likely encoded in "Windows-1252", which includes "ANSI Latin 1" and "Western European (Windows)" encodings.
File Layout Note: If your file's columns layout does not match the GEOLocate specifications, selecting the "custom" option will generate an interface that will allow you to produce a properly laid out version of your uploaded file.
CANCEL
Workbench
Results
Select File
or load an existing file using a retrieval code:
Load
Map data ©2025
Terms


/ blue / general_workshop / se.strickland / Botany2026 / data / 03_georeferencing /
Type
Name
Shortia_galacifolia_Needing_GeoRef_2026_07_22.csv


Home | Web Application | Collaborative Georeferencing | Developer Resources | Education & Outreach | Support and Contacts
GEOLocate Web Application
No location found.
Iceland
Sweden
Norway
Russia
United Kingdom
Poland
Germany
Ukraine
France
Kazakhstan
Mongolia
Spain
Italy
Türkiye
China
South Korea
Japan
Canada
United States
North Atlantic Ocean
Iraq
Iran
Afghanistan
Pakistan
Algeria
Libya
Egypt
Saudi Arabia
India
Thailand
Mexico
Mali
Niger
Chad
Sudan
Nigeria
Ethiopia
Venezuela
Colombia
DRC
Kenya
Tanzania
Indonesia
Papua New Guinea
Peru
Brazil
Bolivia
Angola
Namibia
Botswana
Madagascar
Indian Ocean
South Pacific Ocean
South Atlantic Ocean
Chile
Australia
South Africa
Argentina
New Zealand
2000 km
1000 mi
Google
Off map.
Southern Ocean
Map data ©2025
Terms
Workbench
No location found
Show 8 entries
Page Georeference
Georeference
Options
Correct
Draw polygon
Place marker
Measure
locality string
country
state
county
latitude
longitude
correction status
locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, l
United States of America
South Carolina
NA
NA
locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, l
United States of America
South Carolina
NA
NA
locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, l
United States of America
South Carolina
NA
NA
locality: NA, occurrenceRemarks: Confirmed by AL State Botanist. State record fo
NA
NA
NA
NA
NA
locality: NA, occurrenceRemarks: Oconee Bells. 300+ clone continually spreading
United States of America
South Carolina
NA
NA
locality: NA, occurrenceRemarks: Leaves evergreen; stoloniferous. Petals clear w
United States of America
North Carolina
NA
NA
locality: Fuller Tract, upland woods to ravines., occurrenceRemarks: Persistent
United States of America
Kentucky
NA
NA
locality: NA, occurrenceRemarks: Common Name: "Oconee Bells"; Flowers white, ver
United States of America
South Carolina
NA
NA
Search:
File management
Shortia_galacifolia_Needing_GeoRef_2...
of 561 entries

Home | Web Application | Collaborative Georeferencing | Developer Resources | Education & Outreach | Support and Contacts

GEOLocate Web Application

No location found.



Workbench | No location found

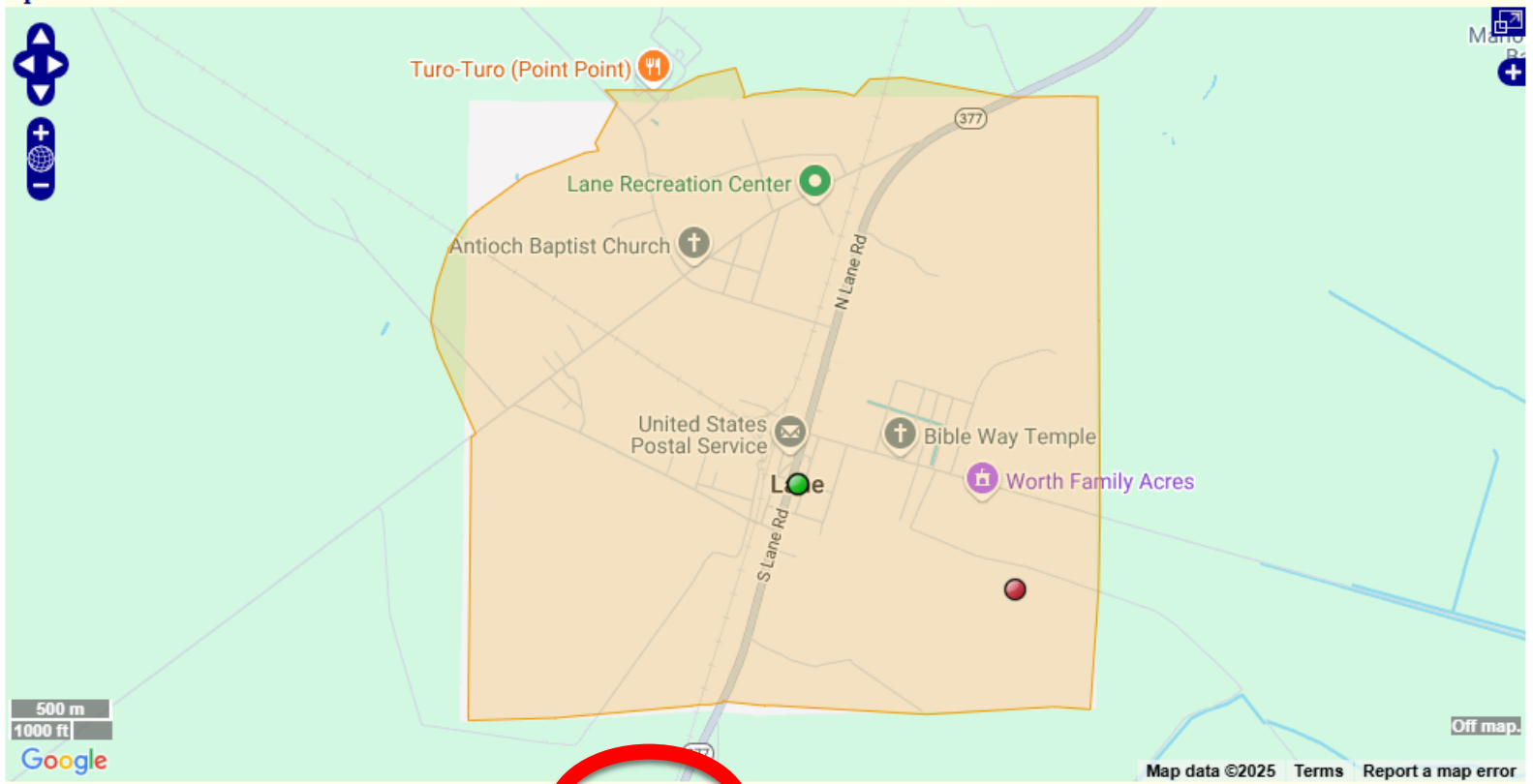
Show 8 entries | Page Georeference | Georeference | Options | Correct | Draw polygon | Place marker | Measure

locality string	country	state	county	latitude	longitude	correction status
locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, I	United States of America	South Carolina	Oconee	33.521555	-79.878685	ves
locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, I	United States of America	South Carolina	Oconee	33.521555	-79.878685	o
locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, I	United States of America	South Carolina	Oconee	33.521555	-79.878685	n
locality: NA, occurrenceRemarks: Confirmed by AL State Botanist. State record fo	NA	NA	NA	-26.5	16.4	n
locality: NA, occurrenceRemarks: Oconee Bells. 300+ clone continually spreading	United States of America	South Carolina	Oconee	34.7667	-83.0833	n
locality: NA, occurrenceRemarks: Leaves evergreen; stoloniferous. Petals clear w	United States of America	North Carolina	McDowell	35.35	-82.35	n
locality: Fuller Tract, upland woods to ravines., occurrenceRemarks: Persistent	United States of America	Kentucky	Powell	37.635654	-82.652656	o
locality: NA, occurrenceRemarks: Common Name: "Oconee Bells"; Flowers white, ver	United States of America	South Carolina	Pickens	34.7667	-83.0833	no

Search: | File management | Shortia_galacifolia_Needing_GeoRef_2... | Showing 1 to 8 of 561 entries

GEOLocate Web Application

2 possible locations found.

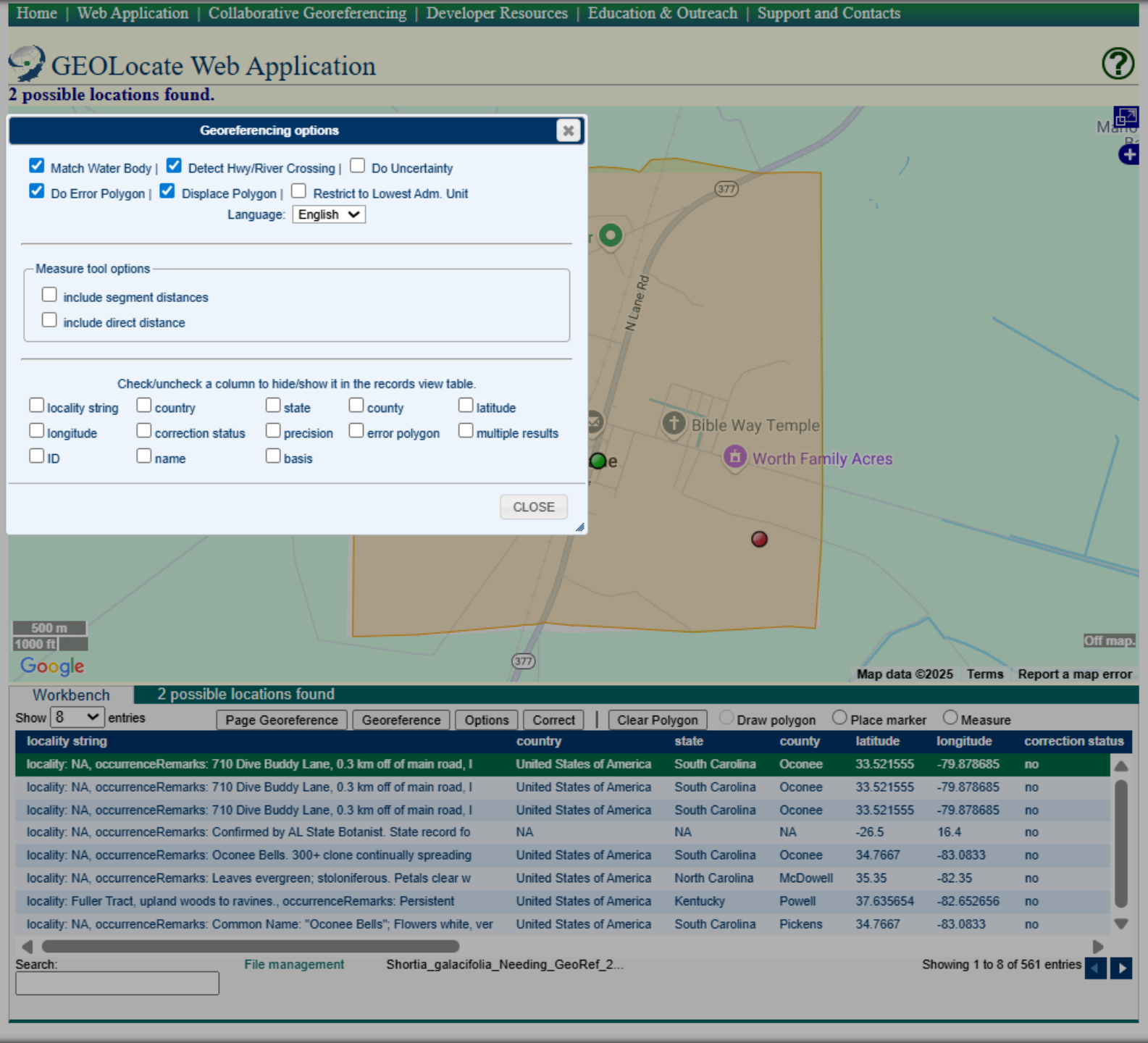


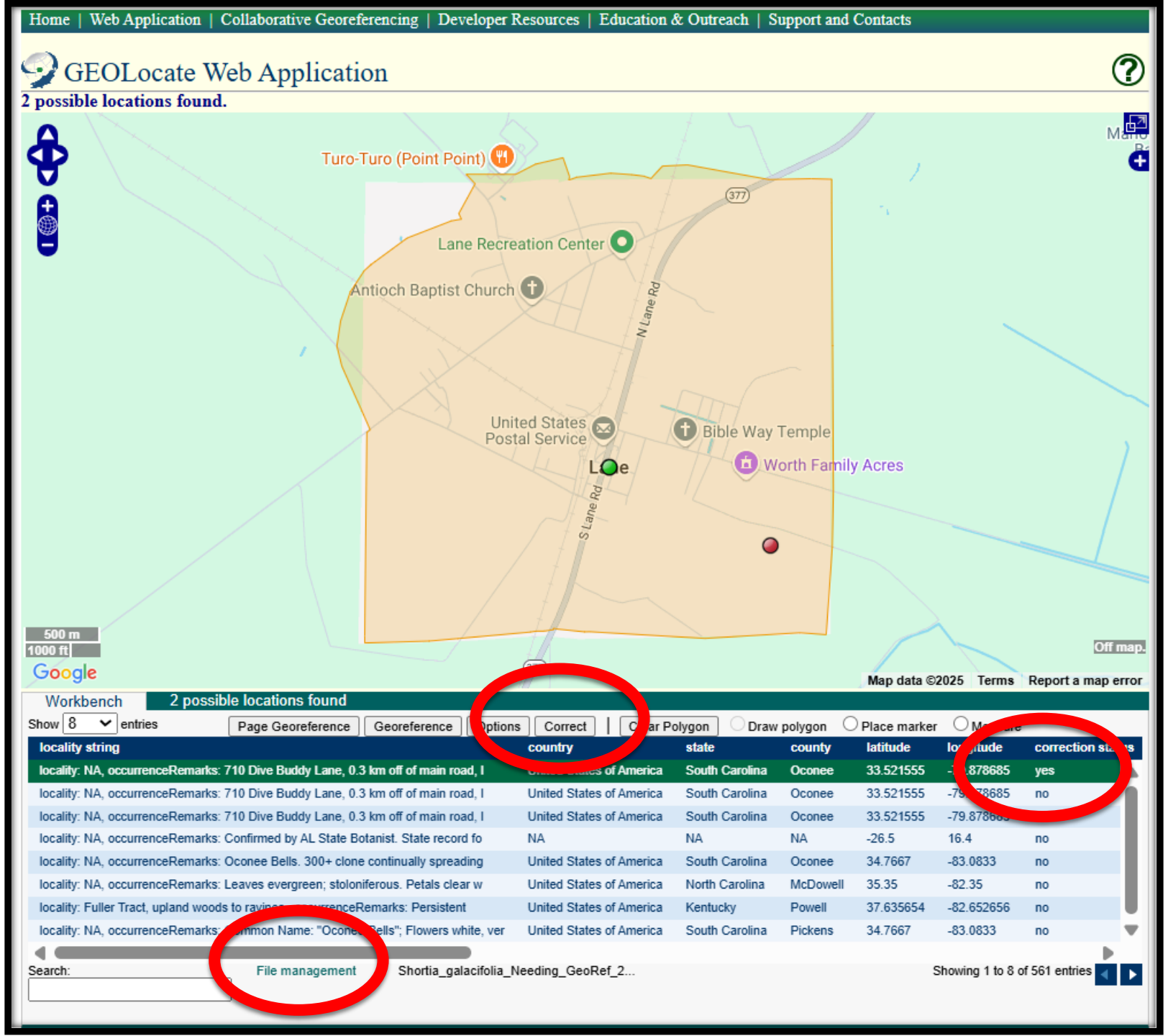
Workbench | 2 possible locations found

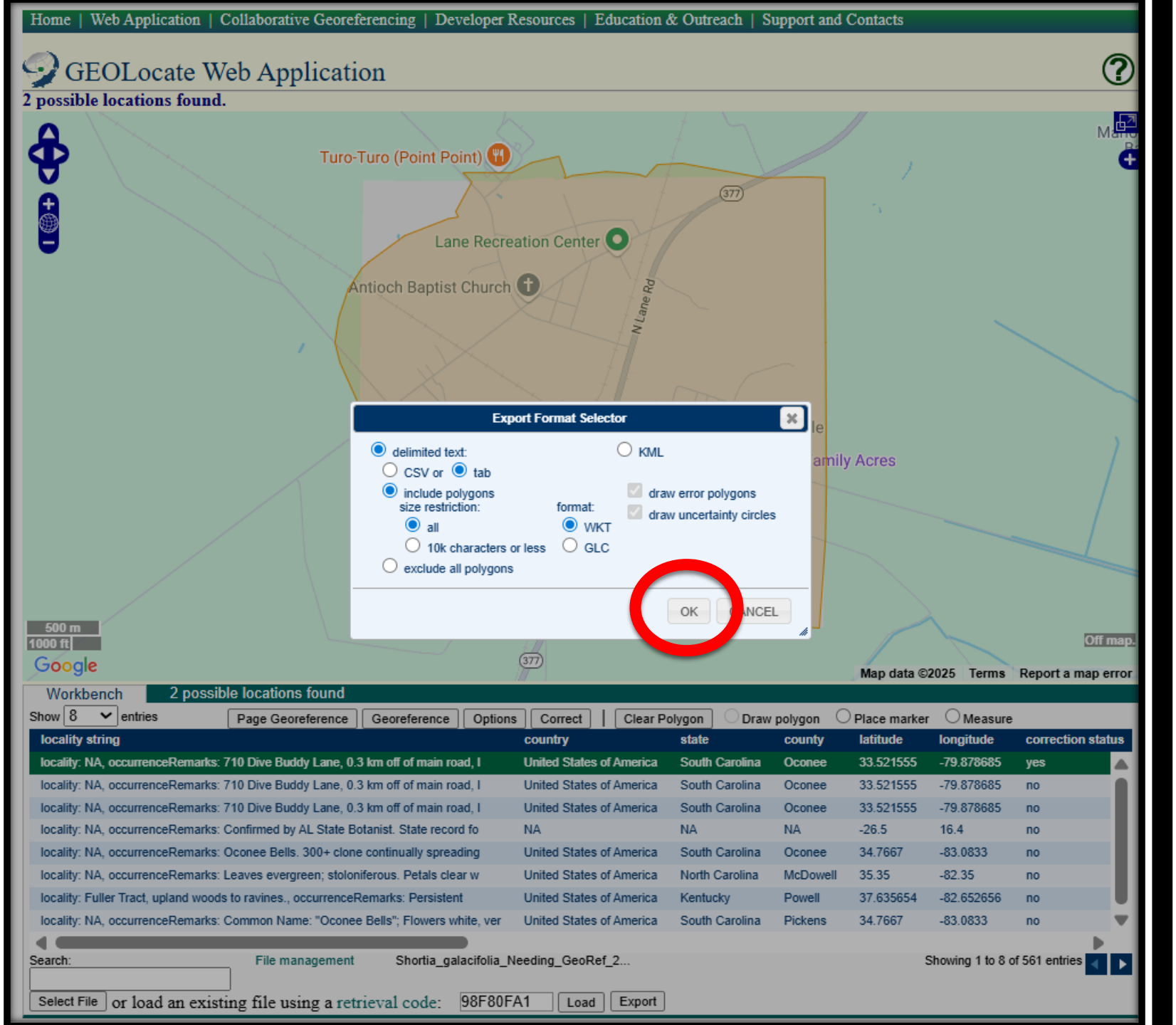
Show 8 entries | Page Georeference | Georeference | Options | Connect | Clear Polygon | Draw polygon | Place marker | Measure

locality string	country	state	county	latitude	longitude	correction status
locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, l	United States of America	South Carolina	Oconee	33.521555	-79.878685	no
locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, l	United States of America	South Carolina	Oconee	33.521555	-79.878685	no
locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, l	United States of America	South Carolina	Oconee	33.521555	-79.878685	no
locality: NA, occurrenceRemarks: Confirmed by AL State Botanist. State record fo	NA	NA	NA	-26.5	16.4	no
locality: NA, occurrenceRemarks: Oconee Bells. 300+ clone continually spreading	United States of America	South Carolina	Oconee	34.7667	-83.0833	no
locality: NA, occurrenceRemarks: Leaves evergreen; stoloniferous. Petals clear w	United States of America	North Carolina	McDowell	35.35	-82.35	no
locality: Fuller Tract, upland woods to ravines., occurrenceRemarks: Persistent	United States of America	Kentucky	Powell	37.635654	-82.652656	no
locality: NA, occurrenceRemarks: Common Name: "Oconee Bells"; Flowers white, ver	United States of America	South Carolina	Pickens	34.7667	-83.0833	no

Search: | File management | Shortia_galacifolia_Needing_GeoRef_2... | Showing 1 to 8 of 561 entries

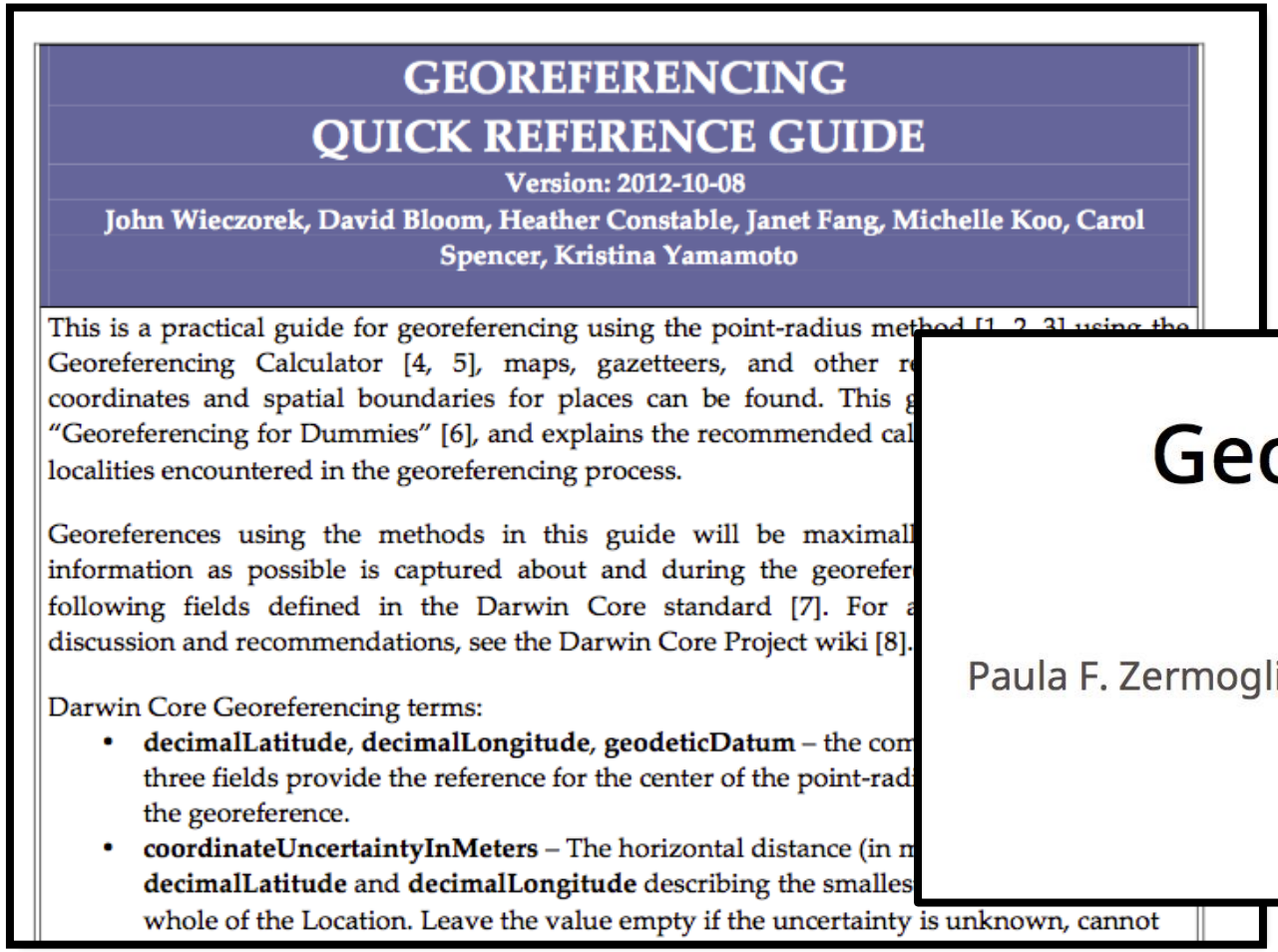

Home | Web Application | Collaborative Georeferencing | Developer Resources | Education & Outreach | Support and Contacts
GEOLocate Web Application
2 possible locations found.
Georeferencing options
Match Water Body | Detect Hwy/River Crossing | Do Uncertainty
Do Error Polygon | Displace Polygon | Restrict to Lowest Adm. Unit
Language: English
Measure tool options
include segment distances
include direct distance
Check/uncheck a column to hide/show it in the records view table.
locality string country state county latitude
longitude correction status precision error polygon multiple results
ID name basis
CLOSE
N Lane Rd
377
Bible Way Temple
Worth Family Acres
500 m
1000 ft
Google
Off map.
Map data ©2025 Terms Report a map error
Workbench
2 possible locations found
Show 8 entries
Page Georeference
Georeference
Options
Correct
Clear Polygon
Draw polygon
Place marker
Measure
locality string | country | state | county | latitude | longitude | correction status
locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, l | United States of America | South Carolina | Oconee | 33.521555 | -79.878685 | no
locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, l | United States of America | South Carolina | Oconee | 33.521555 | -79.878685 | no
locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, l | United States of America | South Carolina | Oconee | 33.521555 | -79.878685 | no
locality: NA, occurrenceRemarks: Confirmed by AL State Botanist. State record fo | NA | NA | NA | -26.5 | 16.4 | no
locality: NA, occurrenceRemarks: Oconee Bells. 300+ clone continually spreading | United States of America | South Carolina | Oconee | 34.7667 | -83.0833 | no
locality: NA, occurrenceRemarks: Leaves evergreen; stoloniferous. Petals clear w | United States of America | North Carolina | McDowell | 35.35 | -82.35 | no
locality: Fuller Tract, upland woods to ravines., occurrenceRemarks: Persistent | United States of America | Kentucky | Powell | 37.635654 | -82.652656 | no
locality: NA, occurrenceRemarks: Common Name: "Oconee Bells"; Flowers white, ver | United States of America | South Carolina | Pickens | 34.7667 | -83.0833 | no
Search:
File management
Shortia_galacifolia_Needing_GeoRef_2...
Showing 1 to 8 of 561 entries


Home | Web Application | Collaborative Georeferencing | Developer Resources | Education & Outreach | Support and Contacts
GEOLocate Web Application
2 possible locations found.
Turo-Turo (Point Point)
Lane Recreation Center
Antioch Baptist Church
United States Postal Service
Bible Way Temple
Worth Family Acres
N Lane Rd
S Lane Rd
377
500 m
1000 ft
Google
Off map
Map data ©2025 Terms Report a map error
Workbench
2 possible locations found
Show 8 entries
Page Georeference
Georeference
Options
Correct
Draw polygon
Place marker
locality string
country
state
county
latitude
correction status
locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, I
United States of America
South Carolina
Oconee
33.521555
yes
locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, I
United States of America
South Carolina
Oconee
33.521555
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locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, I
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South Carolina
Oconee
33.521555
locality: NA, occurrenceRemarks: Confirmed by AL State Botanist. State record fo
NA
NA
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-26.5
16.4
no
locality: NA, occurrenceRemarks: Oconee Bells. 300+ clone continually spreading
United States of America
South Carolina
Oconee
34.7667
-83.0833
no
locality: NA, occurrenceRemarks: Leaves evergreen; stoloniferous. Petals clear w
United States of America
North Carolina
McDowell
35.35
-82.35
no
locality: Fuller Tract, upland woods to ravine
United States of America
Kentucky
Powell
37.635654
-82.652656
no
locality: NA, occurrenceRemarks:
United States of America
South Carolina
Pickens
34.7667
-83.0833
no
Search:
File management
Shortia_galacifolia_Needing_GeoRef_2...
Showing 1 to 8 of 561 entries


Home | Web Application | Collaborative Georeferencing | Developer Resources | Education & Outreach | Support and Contacts
GEOLocate Web Application
2 possible locations found.
Turo-Turo (Point Point)
Lane Recreation Center
Antioch Baptist Church
N Lane Rd
377
Export Format Selector
delimited text:
CSV or tab
include polygons
size restriction:
all
10k characters or less
exclude all polygons
KML
format:
WKT
GLC
draw error polygons
draw uncertainty circles
OK
CANCEL
500 m
1000 ft
Google
Map data ©2025 Terms Report a map error
Workbench
2 possible locations found
Show 8 entries
Page Georeference
Georeference
Options
Correct
Clear Polygon
Draw polygon
Place marker
Measure
locality string | country | state | county | latitude | longitude | correction status
locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, l | United States of America | South Carolina | Oconee | 33.521555 | -79.878685 | yes
locality: NA, occurrenceRemarks: 710 Dive Buddy Lane, 0.3 km off of main road, l | United States of America | South Carolina | Oconee | 33.521555 | -79.878685 | no
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locality: NA, occurrenceRemarks: Confirmed by AL State Botanist. State record fo | NA | NA | NA | -26.5 | 16.4 | no
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locality: NA, occurrenceRemarks: Leaves evergreen; stoloniferous. Petals clear w | United States of America | North Carolina | McDowell | 35.35 | -82.35 | no
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locality: NA, occurrenceRemarks: Common Name: "Oconee Bells"; Flowers white, ver | United States of America | South Carolina | Pickens | 34.7667 | -83.0833 | no
Search:
File management
Shortia_galacifolia_Needing_GeoRef_2...
Showing 1 to 8 of 561 entries
Select File
or load an existing file using a retrieval code:
98F80FA1
Load
Export

Other Resources

Georeferencing Quick Reference Guide



GEOREFERENCING
QUICK REFERENCE GUIDE

Version: 2012-10-08

John Wieczorek, David Bloom, Heather Constable, Janet Fang, Michelle Koo, Carol Spencer, Kristina Yamamoto

This is a practical guide for georeferencing using the point-radius method [1, 2, 3] using the Georeferencing Calculator [4, 5], maps, gazetteers, and other r coordinates and spatial boundaries for places can be found. This g "Georeferencing for Dummies" [6], and explains the recommended cal localities encountered in the georeferencing process.

Georeferences using the methods in this guide will be maximall information as possible is captured about and during the georefer following fields defined in the Darwin Core standard [7]. For a discussion and recommendations, see the Darwin Core Project wiki [8].

Darwin Core Georeferencing terms:

- **decimalLatitude**, **decimalLongitude**, **geodeticDatum** – the com three fields provide the reference for the center of the point-radi the georeference.
- **coordinateUncertaintyInMeters** – The horizontal distance (in m **decimalLatitude** and **decimalLongitude** describing the smalles whole of the Location. Leave the value empty if the uncertainty is unknown, cannot



Georeferencing Quick Reference Guide

Paula F. Zermoglio, Arthur D. Chapman, John R. Wieczorek, María Celeste Luna, David A. Bloom

Version 7adb86d, 2021-02-01 17:30:10 UTC

https://docs.gbif.org/georeferencing-quick-reference-guide/1.0/en/

Other Resources



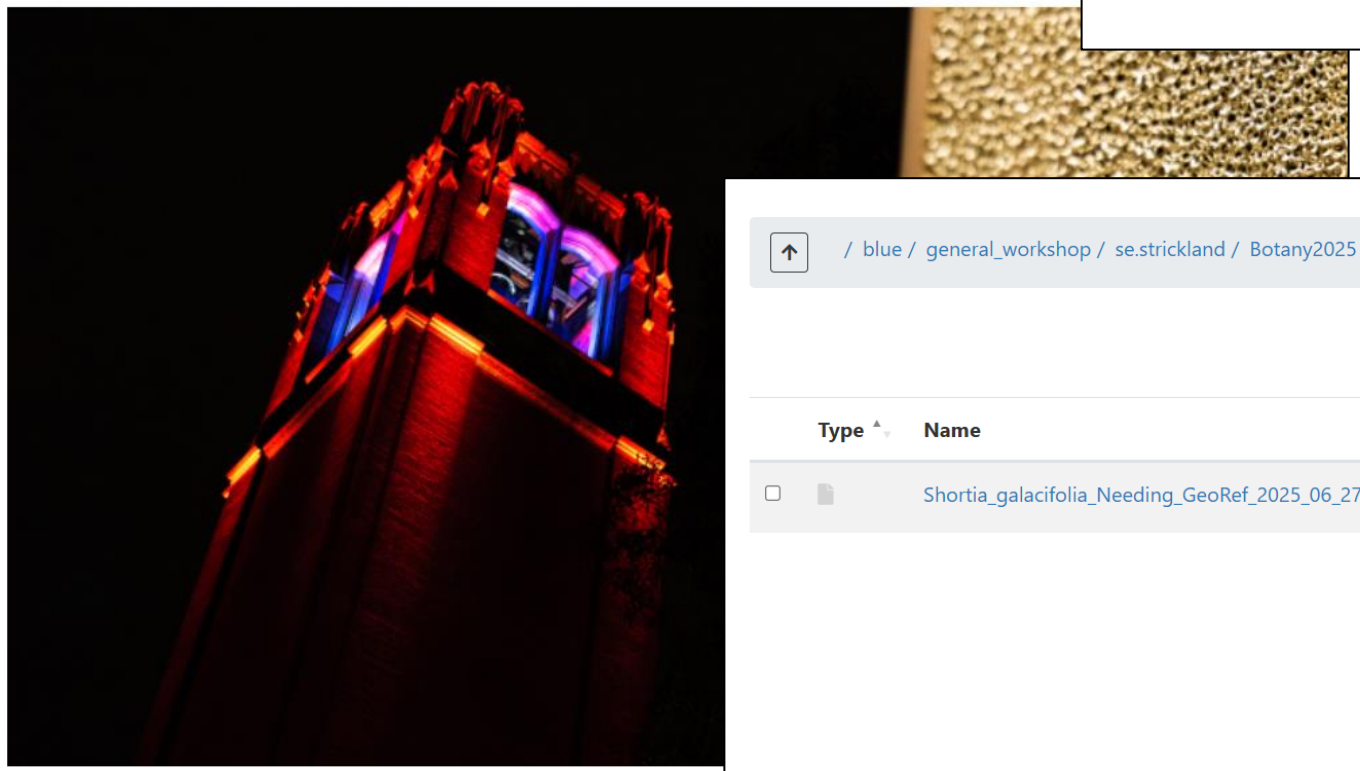
https://www.idigbio.org/wiki/index.php/Georeferencing

Practice

UF Apps Files Jobs Clusters Interactive Apps My Interactive Sessions

Welcome to the G

You are accessing a University of Florida information system and agree to



UF Apps Files Jobs Clusters Interactive Apps My Interactive Sessions Help Logged in as se.strickland Log Out

Open in Terminal | New File | New Directory | Upload | Download | Copy/Move | Delete

Home Directory
/orange/soltis
/blue/soltis
/blue/general_workshop

/ blue / general_workshop / Change directory Copy path

Show Owner/Mode Dotfiles Filter:

Showing 36 rows - 0 rows selected

Type	Name	Size	Modified at	Owner	Mode
	dsoltis	-	7/22/2025 5:46:43 PM	dsoltis	750
	elizabethwhite1	-	7/23/2025 2:09:32 PM	elizabethwhite1	755
	fernandez.se	-	7/23/2025 12:52:54 PM	fernandez.se	755
	guest.100230	-	7/22/2025 5:12:51 PM	guest.100230	700
	guest.100231	-	7/22/2025 5:12:51 PM	guest.100231	700
	guest.100232	-	7/22/2025 5:14:01 PM	guest.100232	700

/ blue / general_workshop / se.strickland / Botany2025 / data / 03_georeferencing / Change directory Copy path

Show Owner/Mode Show Dotfiles Filter:

Showing 1 rows - 0 rows selected

Type	Name	Size	Modified at	Owner	Mode
	Shortia_galacifolia_Needing_GeoRef_2025_06_27.csv	128 KB	7/25/2025 1:51:46 PM	se.strickland	644

View
Edit
Rename
Download
Delete

OnDemand version: 2.0.32